Informational Power and Institutional Design: When a Hegemon May Choose Consensus

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Abstract

When does a powerful state prefer consensus to majority rule in an international organization? I develop a formal model of institutional design in which a hegemon has private information about the value of multilateral cooperation. Under majority rule, weaker states can form winning coalitions without the hegemon, so its private information has limited strategic value. Under unanimity, every agreement requires the hegemon's approval, forcing weaker states to decide how much to offer without knowing how valuable cooperation really is. This uncertainty works in the hegemon's favor: weaker states pay more than necessary to secure agreement, and majority rule eliminates this advantage entirely. The hegemon can further exploit this asymmetry by shaping what weaker states learn before deciding whether to participate. The resulting institutional comparison has a sharp structure: once cooperation is promising enough that weaker states are willing to participate under unanimity, the hegemon strictly prefers it. Consensus, in this account, is not a concession to weaker states but an institutional arrangement that makes informational advantage politically productive—offering a distributive explanation for consensus rules in organizations such as the WTO.

1 Introduction

Most international organizations operate by consensus, even when powerful states drive their creation (Gould 2022). The international arena is where material asymmetries should matter most relative to domestic politics. If institutional rules reflect power, powerful states might be expected to favor arrangements that let them turn that power into a distributive advantage. Why would a powerful state ever prefer consensus over majority rule, where it could use agenda power to impose its preference without any state having veto power?

In bargaining, majority rule allows a powerful actor to form coalitions, exclude rivals, and keep the surplus (Baron and Ferejohn 1989; Kalandrakis 2006; Eraslan and Evdokimov 2019). Consensus does the opposite. By giving every participant a veto, it forces powerful states to surrender the very tools they might value most (Keohane 1984). Yet international organizations, such as the GATT/WTO, are built around consensus (Steinberg 2002; Koremenos, Lipson, and Snidal 2001).

Existing answers do not fully resolve the puzzle. A common response claims that powerful states still influence outcomes through informal agenda power, even under consensus (Steinberg 2002; Stone 2011). Yet this answer is conceptually unsatisfying: under unanimity, all participants are veto players, so informal agenda influence cannot enable exclusion and surplus capture as majority rule does. More broadly, existing accounts either see consensus as constraining powerful states—a mechanism of institutional self-binding that limits the hegemon's own discretion (Keohane 1984; Ikenberry 2001)—or as a concealment for conventional power, where formal equality masks asymmetric influence (Steinberg 2002; Gruber 2000). Neither explains when and why a hegemon would prefer unanimity to a rule that allows exclusion.

This paper theorizes when and why a hegemon may prefer unanimity to majority rule in international organizations. The central claim is that consensus is an institutional technology of power. Using Bayesian persuasion (Kamenica 2019), I show that consensus can make informational power more politically productive than the coalition arithmetic of majority rule. While the multi-receiver persuasion literature has examined information design under unanimity (Bardhi and Guo 2018), and Kim, Kim, and Van Weelden (2025) analyzes Bayesian persuasion in bilateral veto bargaining, neither offers the institutional comparison between voting rules that is central here. Thus, the key institutional choice is between two distinct modes of exercising power: agenda power and informational power.

To formalize the argument, I develop a model in which a hegemon chooses between majority and unanimity. Proposal rights are symmetric under both rules, isolating the effect of the voting rule itself. Before weaker states decide whether to enter, the hegemon privately observes the value of cooperation and can shape their beliefs about it. The central difference is that the two rules reward this informational advantage differently. Under majority rule, weak states can form coalitions without making the hegemon pivotal, eliminating the screening problem. Under unanimity, weak states must include the hegemon in every proposal and decide how much to offer without knowing the probability of acceptance. This screening generates a cutoff at which weak proposers switch from aggressive to conservative treatment of the hegemon, giving the hegemon a strictly higher conditional payoff than under majority at every level of uncertainty. Bayesian persuasion extends this conditional advantage to priors where the institution would not otherwise form under unanimity, but majority may still be preferred when its easier participation conditions are decisive.

The substantive implication is that consensus is most valuable to a hegemon when the prospects of multilateral cooperation are promising enough that weaker states are willing to come to the table. Under those conditions, unanimity amplifies the hegemon's informational advantage. When cooperation looks too uncertain for weaker states to participate at all, majority rule can be preferred—not because it gives the hegemon better bargaining terms, but because it lowers the bar for participation.

The rest of the paper proceeds as follows. Section 2 presents a motivating example that previews the mechanism in a simplified three-player environment. Section 3 presents the general model. Sections 4 and 5 derive the bargaining logic under majority and unanimity, showing that unanimity creates a screening cutoff and a discrete jump in the hegemon's payoff that majority eliminates. Section 6 connects bargaining to entry and persuasion. Section 7 establishes the main result. Section 8 discusses the GATT/WTO application and scope conditions. Section 9 concludes.

2 A Motivating Example

Throughout the paper, I model consensus as unanimity over a binary accept/reject decision. This abstracts from agenda-setting conventions and silence procedures, but captures the central strategic feature: any member can block agreement. Before turning to the general model, consider a simplified special case with three players and one bargaining round to build intuition for the mechanism. The general model (Section 3) extends this to N players, two rounds of Baron-Ferejohn bargaining, and entry costs. There is one hegemon H and two weak states W_1, W_2. The value of cooperation is either $V(0) = 1$ (low) or $V(1) = 2$ (high); H observes the state, the weak states do not. Each weak state's outside option is 0; H's outside option is $\alpha V(\theta)$ with $\alpha = 0.2$. A proposer is selected uniformly at random; to illustrate the screening mechanism, I focus on the case where a weak state is selected and makes a take-it-or-leave-it offer specifying H's share y_H. If H rejects, each player receives their disagreement payoff.

Majority: exclusion eliminates screening. Under majority rule, W_1 and W_2 can pass a proposal without H's vote. The proposing weak state excludes H from the coalition; H captures $0.2V(\theta)$ from its bilateral alternative regardless of what W believes. The hegemon's expected payoff is $0.2 + 0.2\mu$, where $\mu = \Pr(\theta = 1)$—linear in beliefs, with no strategic discontinuity.

Unanimity: inclusion creates screening. Under unanimity, H's vote is required. The proposing weak state must decide how much to offer H without knowing θ:

- *Aggressive offer* ($y_H = 0.2$): type $\theta = 0$ is indifferent and accepts; type $\theta = 1$ rejects (wants 0.4). W gets 0.8 if $\theta = 0$, gets 0 if $\theta = 1$.
- *Conservative offer* ($y_H = 0.4$): both types accept. W gets 0.6 if $\theta = 0$, gets 1.6 if $\theta = 1$.

The weak state prefers the aggressive offer when $0.8(1-\mu) > 0.6(1-\mu)+1.6\mu$, which simplifies to $\mu < 1/9$. The screening cutoff $\mu^* = 1/9$ is the belief at which W switches from aggressive to conservative treatment of H.

The jump and overpayment. Below the cutoff, H's expected payoff is the same as under majority: $0.2 + 0.2\mu$. Above the cutoff, H's expected payoff jumps to 0.4—constant, independent of μ. The source is overpayment: on the conservative branch, type $\theta = 0$ receives 0.4 but would accept 0.2. Weak states cannot avoid this cost because they must buy H's vote without knowing the type. The jump at $\mu^* = 1/9$ is $8/45 \approx 0.18$, or about 16% of expected surplus at the cutoff.

Bayesian persuasion exploits the jump. The jump creates a non-convexity in H's value function that Bayesian persuasion can exploit. Suppose the prior is $p < 1/9$. Without persuasion, H's payoff under both rules is $0.2 + 0.2p$. But under unanimity, H can design a public signal that pushes W's posterior to $1/9$ with probability $9p$ (inducing conservative behavior and payoff 0.4) and to 0 with probability $1 - 9p$ (payoff 0.2). The resulting expected payoff is $0.2 + 1.8p$, strictly exceeding the majority payoff of $0.2 + 0.2p$ for every $p > 0$. Under majority, the payoff is already linear, so concavification adds nothing. This is the central asymmetry: unanimity creates a non-convexity that makes information strategically productive; majority does not.

Preview. The general model (Section 3 onwards) enriches this logic with N players, random proposer, two-round Baron-Ferejohn bargaining with discounting ($\beta < 1$), and an explicit entry stage with cost c. The richer environment produces two screening cutoffs (one per round), a richer non-convexity structure, and an entry threshold that creates a second channel through which the voting rule matters. Theorem 1 shows that the pattern here—unanimity dominates whenever the institution is viable—generalizes beyond this example, and Theorem 2 establishes a single-crossing property for the full institutional comparison.

3 The Model

Definition 1 (Institutional Design Game)**.** *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $N-1$ weak states $W_1, \dots, W_{N-1}$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior $\Pr(\theta = 1) = p \in (0, 1)$.*
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ; the weak states do not.*
- *Disagreement payoffs: each W receives 0;*[1] *H receives $\alpha V(\theta)$, with $\alpha \in (0, 1/r)$.*[2]
- *The common discount factor is $\beta \in (0, 1)$.*

[1] The normalization $d_W = 0$ is without loss of generality: it requires only that weak states' outside options are symmetric and strictly lower than the hegemon's. The pie $V(\theta)$ is then interpreted as the surplus above weak states' common outside option.

[2] The proportional form $d_H = \alpha V(\theta)$ captures bilateral alternatives that scale with the value of multilateral cooperation—the same factors that make multilateral cooperation valuable also improve the hegemon's outside option. The restriction $\alpha < 1/r$ ensures bilateral alternatives are strictly dominated by multilateral cooperation for all states of the world.

- *Each weak state that joins the institution pays an entry cost $c > 0$.*
- *The voting rule is $R \in \{M, U\}$: under majority (M), decisions pass with $q = \lfloor N/2 \rfloor + 1$ votes; under unanimity (U), decisions require all N votes.*[3] *Both rules have symmetric proposal rights (each player recognized with probability $1/N$).*

Let $V_e(\mu) \equiv 1 + \mu(r-1)$ denote the expected value of cooperation at posterior belief μ, and define $x \equiv (N-1)\alpha r$ as a notational shorthand.

The two-round bargaining structure follows the Baron-Ferejohn framework that is standard in legislative bargaining. A single round would reduce the game to a one-shot ultimatum, and one might question whether the screening mechanism depends on that special structure. Two rounds ensure that rejection in Round 1 leads to a continuation game—not immediate disagreement—so the hegemon's reservation value in Round 1 reflects a credible outside option within the institution. The second round also generates off-path belief updating: when the high type rejects an aggressive offer in Round 1, weak states learn $\theta = 1$ with certainty, which pins down the Round 2 offers and, by backward induction, the Round 1 screening cutoff. Beyond robustness, the two-round structure generates analytically richer results: the R1 screening cutoff is independent of α (Proposition 2), and the two screening cutoffs (one per round) create a richer non-convexity that Bayesian persuasion can exploit. The motivating example (Section 2) shows that the mechanism may operate in one round; the two-round structure confirms that it survives in the richer Baron-Ferejohn environment.

The game has three stages.

1. **Stage 0 (Institutional choice).** The hegemon chooses the voting rule $R \in \{M, U\}$.
2. **Stage 1 (Information and entry).** The hegemon commits to a public signal structure $\pi : \{0, 1\} \to \Delta(S)$. Nature draws θ; the hegemon observes θ and the signal realization $s \in S$ is generated according to π. Weak states observe s, update to posterior $\mu = \Pr(\theta = 1 \mid s)$, and simultaneously decide whether to enter (paying cost c). The hegemon participates automatically—it has chosen the institution at Stage 0 and faces no entry cost. The institu-

[3] In this game, each player's action is binary: accept or reject. There is no abstention or silence option. The institutional distinction between consensus—where silence is treated as tacit approval—and formal unanimity—where explicit approval is required—therefore does not arise: with binary actions {accept, reject}, the two rules are equivalent. I use the terms interchangeably. In practice, "consensus" at the WTO refers to the norm of not calling a formal vote, which approximates unanimity when any member can block by objecting.

tion forms if and only if all $N-1$ weak states enter; partial entry is not permitted.[4] Since weak states are ex ante identical and observe the same public signal, I restrict attention to the symmetric entry equilibrium: all $N-1$ weak states enter if and only if the expected equilibrium payoff satisfies $V_W^{R1}(\mu, R) \geq c$; otherwise no institution forms and each player receives their disagreement payoff.

3. **Stage 2 (Bargaining).** If the institution forms, members play a two-round Baron-Ferejohn bargaining game over a pie of size $V(\theta)$:
   - **Round 1**: a proposer is selected uniformly at random (probability $1/N$). The proposer offers a division. All members vote according to rule R. If accepted, payoffs are realized. If rejected, the game proceeds to Round 2 with payoffs discounted by β.
   - **Round 2 (terminal)**: a new proposer is selected uniformly at random. The proposer offers a division. All members vote. If accepted, payoffs are realized. If rejected, each player receives their disagreement payoff.

The solution concept is Perfect Bayesian Equilibrium (PBE). Strategies are sequentially rational given beliefs, and beliefs are updated by Bayes' rule on the equilibrium path. Off-path beliefs satisfy sequential rationality.[5]

Throughout the game, the hegemon's information sets are singletons: H knows θ at every decision node. Each weak state's information sets pool the two branches of θ: at any decision node, W_i knows the public signal realization and the history of proposals and responses, but not θ. After the public signal, weak states hold posterior μ over θ. This asymmetry is what generates the screening problem when a weak state proposes under unanimity.

Figures 1 and 2 present the extensive form of the game: Figure~1 covers institutional choice, Bayesian persuasion, and entry (Stages 0–1); Figure~2 details the bargaining subgame under unanimity (Stage 2).

[4] The all-or-nothing entry technology is a simplification that avoids a separate extensive form for coalition formation at the entry stage. It is appropriate when the institution requires broad membership to function (as in a multilateral trade round) or when the voting rule is defined over a fixed membership. The parameter N should be read as the size of the institution the hegemon proposes to create, not the total number of states in the international system. The hegemon invites a specific group of $N-1$ weak states; the entry decision is whether to join this particular institution.

[5] Tie-breaking convention: when H is indifferent between accepting and rejecting an offer (offer equals reservation value), H accepts.

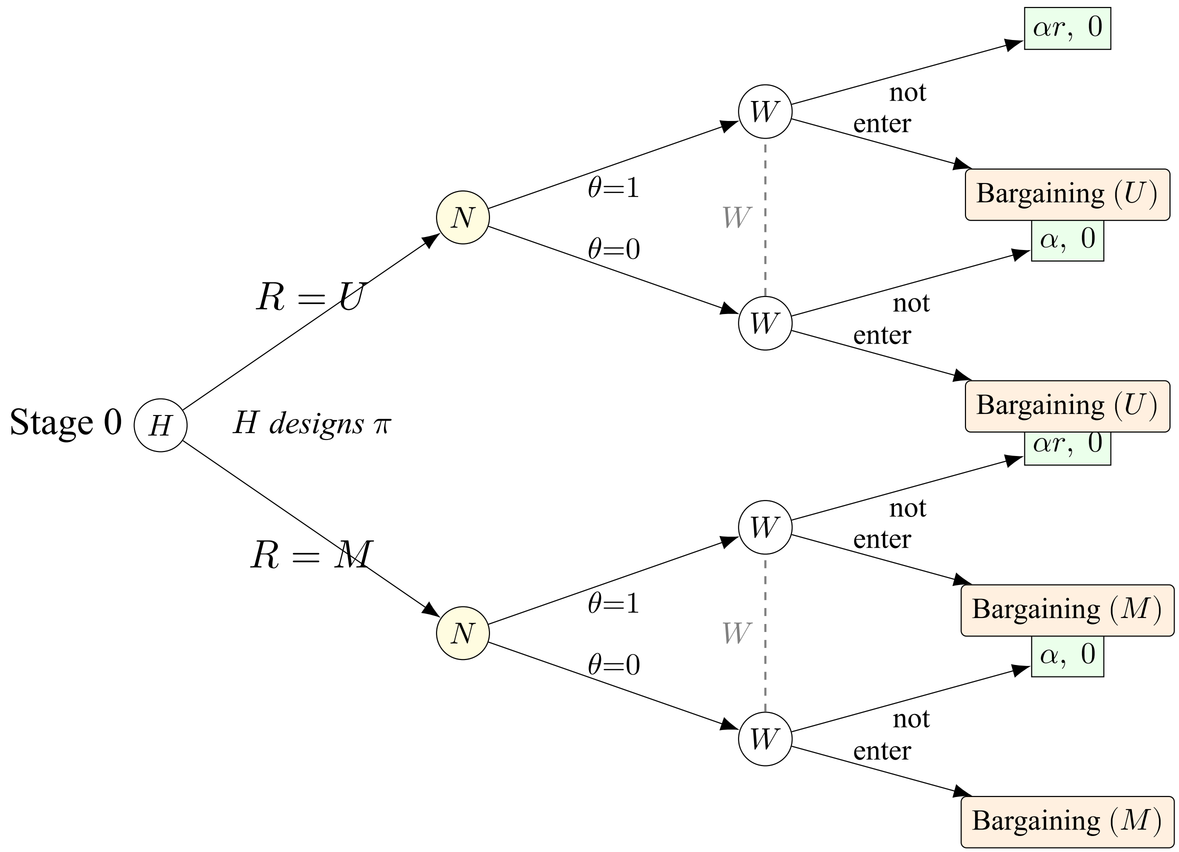

Figure 1: Extensive form: Stages 0–1 (institutional choice, information design, and entry). At Stage 0, H selects voting rule $R \in \{M, U\}$ and commits to a Bayesian persuasion signal π. Nature draws $\theta \in \{0, 1\}$; weak states observe signal realization s, form posterior $\mu = \Pr(\theta{=}1 \mid s)$, and decide entry (cost c). Dashed lines: W's information sets (W does not observe θ). Non-entry payoffs: $(u_H, u_W) = (\alpha V(\theta),\ 0)$.

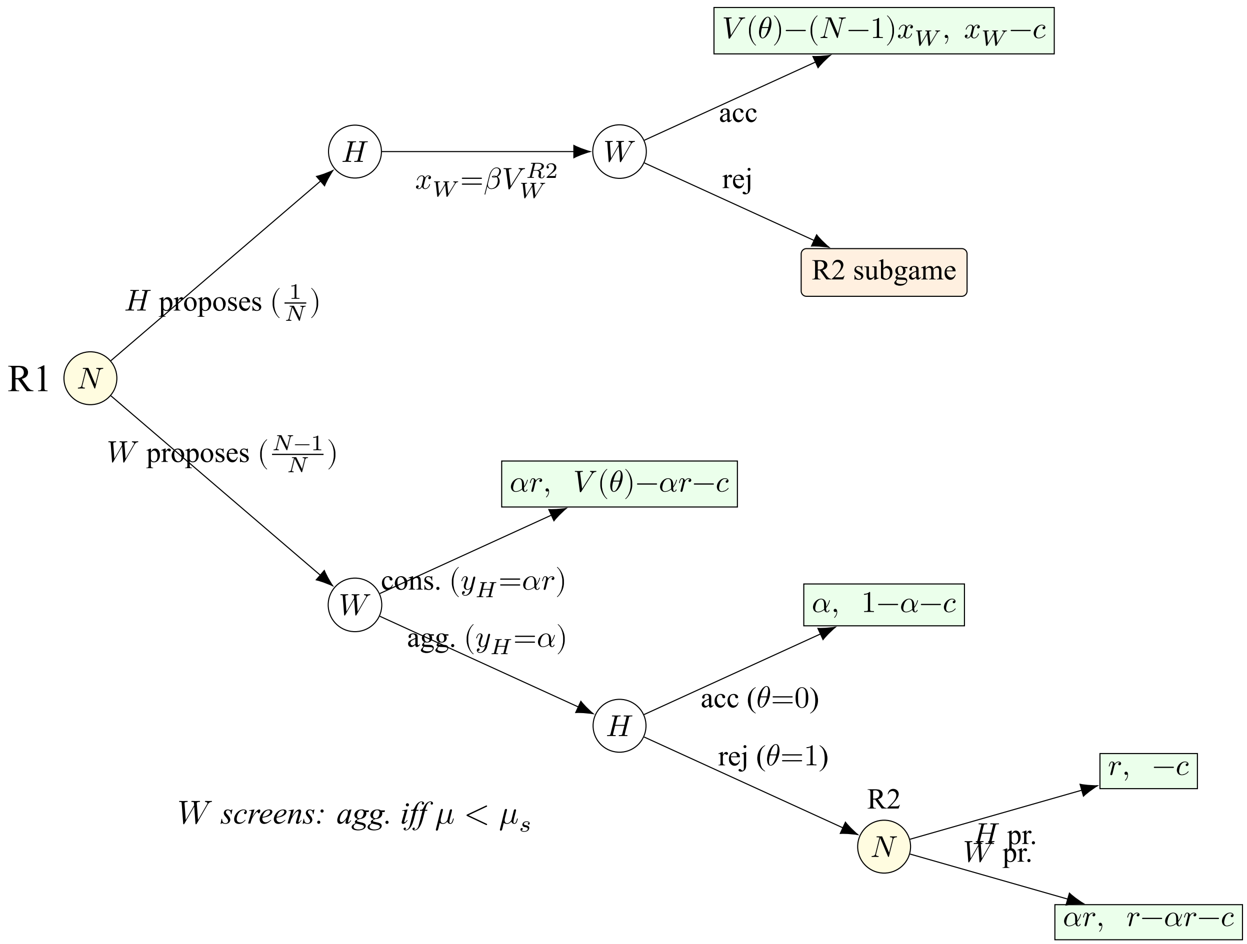

Figure 2: Extensive form: Stage 2 (bargaining under unanimity, Rounds 1–2). Payoffs are (u_H, u_W). A proposer is selected uniformly $(1/N)$. When W proposes, W chooses between an aggressive offer that only $\theta=0$ accepts and a conservative offer that both types accept. Under the aggressive offer, $\theta=1$ rejects, revealing the state and leading to R2 (payoffs discounted by β). In R2, W knows $\theta=1$ (complete information); W offers αr, H offers 0; disagreement yields $(\alpha r, -c)$. When H proposes in R1, H offers βV_W^{R2} to each W; W accepts in equilibrium. Offer labels and terminal payoffs show R2 values (where H's reservation is $\alpha V(\theta)$) for readability; exact R1 offer levels are $\beta V_H^{R2}(\theta=0, \mu'=1)$ (aggressive) and $\beta V_H^{R2}(\theta=1)$ (conservative), derived in Appendix A.3. Under majority rule, H's vote is not pivotal, eliminating the screening mechanism (Proposition 1).

3.1 Preview of the main result

The paper's main results come in two parts. Theorem 1 establishes that whenever the institution can form under unanimity, the hegemon strictly prefers it: the screening mechanism gives unanimity a conditional payoff advantage that majority's linearity cannot overcome. Theorem 2 then characterizes the full institutional comparison: the set of priors at which unanimity dominates is an upper interval, with majority dominating only at pessimistic priors where the entry margin is binding. The mechanism rests on three building blocks developed in Sections 4–@ref(entry_bp): (1) majority produces no screening, so the hegemon's payoff is linear in beliefs (Section 4); (2) unanimity forces weak states to screen the hegemon's type, creating a discrete upward jump in the hegemon's expected payoff (Section 5); and (3) this conditional advantage persists after Bayesian persuasion, which extends unanimity's dominance to priors below the entry threshold (Section @ref(entry_bp)). The following sections develop each building block before assembling them into the institutional comparison in Section 7.

4 Majority Rule: Linear Bargaining Without Screening

Under majority rule, a proposal passes with $q = \lfloor N/2 \rfloor + 1$ votes. The key structural feature is that weak states can form a winning coalition without including the hegemon. When a weak state proposes, it assembles a majority from the other weak states alone; the hegemon, excluded from the coalition, captures its bilateral alternative $\alpha V(\theta)$ regardless of what weak states believe about the state of the world.

Proposition 1 (Majority rule produces no screening)**.** *Under majority rule with symmetric proposal rights, the hegemon's expected continuation value $E_\theta[V_H^{R1}(\theta, \mu, M)]$ is affine in posterior beliefs μ. There is no screening cutoff.*

Proof. See Appendix B.1. □

Because the hegemon's vote is never needed, weak states never face a choice between offers that depend on inferring the hegemon's type. The hegemon's private information affects the value of the agreement but creates no strategic discontinuity. Majority transforms bargaining into coalition arithmetic: the proposer buys the cheapest votes, and the hegemon is simply paid its outside option. Bayesian persuasion under majority therefore operates only through the entry channel—it can induce weak states to participate, but it cannot extract screening rents from the bargaining process itself.

5 Consensus: Screening Through Pivotal Inclusion

Under unanimity, every proposal requires the approval of all members, including the hegemon. When a weak state proposes, it must secure H's vote without knowing whether the pie is high or low. As in the motivating example (Section 2), this creates a screening problem: the proposer must choose between an aggressive offer that only the low type accepts and a conservative offer that both types accept. The conservative offer overpays the low type—the price of uncertainty.

This screening logic operates in both bargaining rounds. In each round, the weak proposer's indifference between aggressive and conservative offers defines a screening cutoff: a belief threshold above which the proposer switches from aggressive to conservative treatment. The R2 cutoff has a closed form in all cases; the R1 cutoff has a closed form when the hegemon's outside option is moderate (Appendix A).

Proposition 2 (R1 screening cutoff under consensus)**.** *Under unanimity, there exists a unique screening cutoff $\mu_s^{R1} \in (0,1)$ such that the weak proposer prefers the aggressive offer for $\mu < \mu_s^{R1}$ and the conservative offer for $\mu > \mu_s^{R1}$. When $\alpha < \bar{\alpha}(r, \beta, N)$, the cutoff takes the closed form:*

$$\mu_s^{R1} = \frac{\phi - \sqrt{\phi^2 - 4(N-2)}}{2(N-2)}, \qquad \phi \equiv \frac{rN - \beta(N-1+r)}{\beta(r-1)}, \tag{1}$$

and is independent of α. The regime threshold is

$$\bar{\alpha}(r, \beta, N) \equiv \frac{\mu_s^{R1} \cdot r}{r - 1 + \mu_s^{R1}}, \tag{2}$$

where μ_s^{R1} is evaluated from (1)*. For $\beta = 1$, the cutoff simplifies to $\mu_s^{R1} = 1/(N-2)$.*[6]

[6]When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch and depends on α (Appendix A.5). The cutoff remains unique and the screening jump remains positive. The proof of Lemma 1 (Appendix B.5) covers both regimes explicitly: the payoff decomposition is additive in the R1 and R2 corrections regardless of the relative ordering of μ_s^{R1} and μ_s^{R2},

Proof. See Appendix B.2. □

The independence of the screening cutoff from the hegemon's outside option strength—which holds when $\alpha < \bar{\alpha}$—is substantively important. What determines how aggressively weak states treat the hegemon is not how valuable bilateral alternatives are, but how consequential the hegemon's private information is for the value of cooperation. The cutoff also depends on bargaining patience and the number of states that must agree.

Proposition 3 (Screening creates an informational rent)**.** *Under unanimity, the hegemon's expected continuation payoff in R1 has an upward jump at* μ_s^{R1}*:*

$$\textit{Jump} = (1 - \mu_s^{R1}) \cdot \frac{(N-1)\beta(r-1)}{N^2} > 0. \tag{3}$$

Proof. See Appendix B.3. □

At the screening cutoff, weak proposers switch from treating the hegemon as the low type to hedging against the possibility of the high type. The conservative offer pays the low type as if it could be the high type, creating a discrete upward shift in the hegemon's expected payoff. The screening rent is largest when the hegemon's private information is most consequential for the value of cooperation (large r), when bargaining is patient enough to make the continuation threat credible (large β), and in moderately-sized organizations—large enough that the hegemon's vote matters, but not so large that the cost of unanimous agreement dilutes the per-member advantage ($(N-1)/N^2$ peaks at intermediate N).

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and the endpoint argument establishes $D(\mu) > 0$ on all branches. Theorems 1–2 then follow from Lemma 1 without further dependence on the regime.

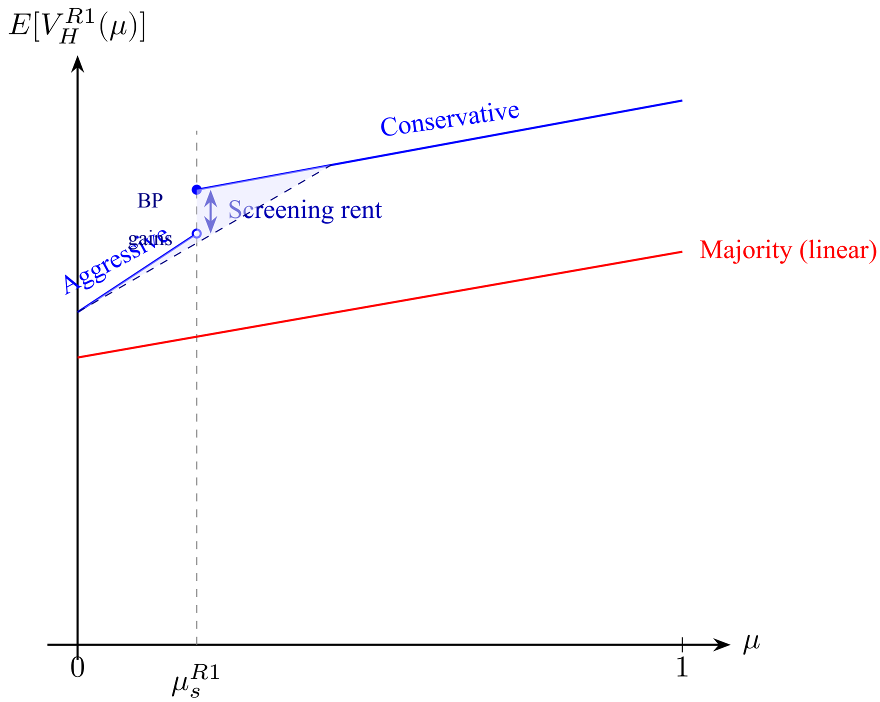

Figure 3: Schematic of the hegemon's expected R1 payoff as a function of posterior belief μ, based on Example 1 parameters ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$). Under majority (red), the payoff is linear in μ with no discontinuity. Under unanimity (blue), the payoff has two branches separated by an upward jump at the screening cutoff $\mu_s^{R1} \approx 0.197$: to the left, weak proposers play aggressively; to the right, they switch to conservative offers that overpay the low type. The shaded region shows where the concave envelope (dashed) lies above the value function, creating gains from Bayesian persuasion.

Example 1 (Screening jump: concrete magnitudes)**.** *Let $N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$. The R1 screening cutoff is $\mu_s^{R1} \approx 0.197$. At beliefs just below the cutoff, weak proposers play aggressively and the hegemon's expected payoff is $E[V_H^{R1}] \approx 0.544$. Just above the cutoff, the payoff jumps to ≈ 0.602—an increase of about 5.3% of expected surplus. Under majority at the same belief, $E[V_H^{R1}] \approx 0.428$, with no jump anywhere. The screening mechanism gives the hegemon 27% more than majority on the aggressive branch and 41% more on the conservative branch. The discrete gap between branches is the non-convexity that Bayesian persuasion exploits.*

6 Entry and Bayesian Persuasion

A weak state enters the institution if its expected bargaining payoff exceeds the entry cost $c > 0$.[7] Because unanimity gives the hegemon a higher payoff conditional on participation (as shown in Section 7 below), it leaves less surplus for weak states, making entry harder: $\tau(U) \geq \tau(M)$.

When the institution does not form, the hegemon receives $\alpha V_e(\mu)$ from its bilateral alternative. Since this outside-option payoff is affine in μ and common to both rules, the institutional comparison depends only on the gain from bargaining relative to this baseline.

Definition 2 (Net gain function)**.** *The hegemon's* net gain *from the institution under rule R at posterior μ is*

$$v(\mu, R) \equiv \begin{cases} E_\theta[V_H^{R1}(\theta, \mu, R)] - \alpha V_e(\mu) & \textit{if weak states enter,} \\ 0 & \textit{otherwise.} \end{cases}$$

Under majority, $v(\mu, M)$ is affine above the entry threshold—a single non-convexity. Under unanimity, the screening jump at μ_s^{R1} creates an additional non-convexity.

Proposition 4 (Unanimity creates an additional persuasion opportunity)**.** *The value function $v(\mu, U)$ has a screening non-convexity at μ_s^{R1} that is absent in $v(\mu, M)$.*

Proof. See Appendix B.4. □

By the standard Bayesian persuasion result (Kamenica and Gentzkow 2011), the hegemon's optimal payoff under rule R is $\Pi_H^*(R, p) = \alpha V_e(p) + \operatorname{cav} v(p, R)$: the outside option plus the concavified net gain from the institution. Since $\alpha V_e(p)$ is affine and common to both rules, the institutional comparison reduces to comparing $\operatorname{cav} v(p, U)$ versus $\operatorname{cav} v(p, M)$. Under majority, the conditional payoff is linear in beliefs (Proposition 1), so concavification operates only through the entry threshold. Under unanimity, concavification can additionally exploit the screening jump,

[7] The entry cost c may represent fixed costs of negotiating accession, per-round costs of preparing delegations, or political costs of committing to multilateral disciplines. In established organizations, c is best interpreted as the cost of *substantive engagement*: allocating scarce diplomatic and analytical resources to a particular negotiating round. Since $V(\theta)$ is normalized independently of N, per-capita payoffs scale as $O(1/N)$. For large-N applications, c should be read as a per-capita cost: writing $c = \tilde{c}/N$, the entry condition becomes $N \cdot V_W \geq \tilde{c}$, where $N \cdot V_W$ is $O(1)$. All equilibrium cutoffs (μ_s^{R1}, μ_s^{R2}, α^*, p^*) are homogeneous of degree zero in the surplus and thus independent of this normalization.

inducing posteriors that straddle μ_s^{R1} and place mass on both sides of the discontinuity. The distinctive contribution of Bayesian persuasion under unanimity is not to create the conditional payoff advantage—that comes from screening (Lemma 1)—but to extend it to priors below the entry threshold, where the institution would not otherwise form. Figure 4 illustrates both net-gain functions and their concavifications.

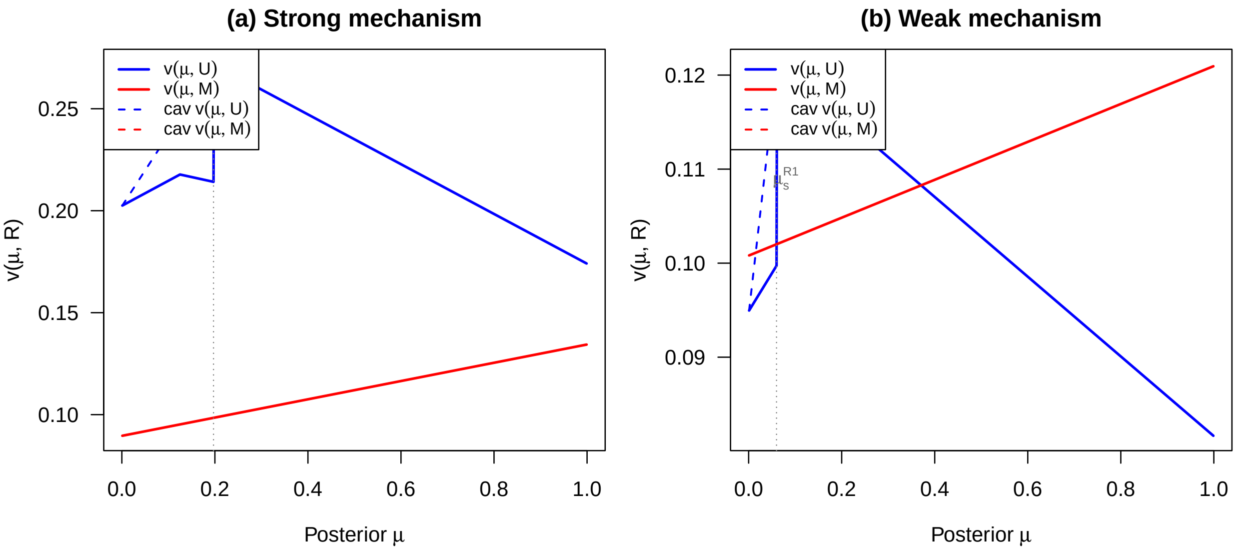

Figure 4: Net-gain functions $v(\mu, R)$ and their concavifications under majority (red) and unanimity (blue). The net gain is the hegemon's expected bargaining payoff minus the bilateral outside option $\alpha V_e(\mu)$; it equals zero when the institution does not form. Dashed lines show concave envelopes. Left panel (a): baseline parameters ($N = 5$, $r = 1.5$, $\alpha = 0.3$, $\beta = 0.9$, $c = 0.1$) where the screening jump under unanimity creates a large concavification gain. Right panel (b): parameters near the boundary of the unanimity-preferred region ($N = 5, r = 1.2, \alpha = 0.3, \beta = 0.7, c = 0.1$) where the mechanism is weaker and the concavification gains are smaller.

7 Institutional Choice

This section establishes the paper's main result: conditions under which the hegemon prefers unanimity.

7.1 Conditional payoff comparison

Before turning to the full institutional comparison with persuasion, I establish a result that is central to the analysis: conditional on the institution forming, unanimity gives the hegemon a strictly higher expected continuation payoff than majority for all interior beliefs.

Define the *conditional dominance threshold*

$$\alpha^*(N,\beta) \equiv \frac{\beta(q-1)}{N(N-1)-\beta(N^2-N-q+1)}, \qquad q = \lfloor N/2 \rfloor + 1. \tag{4}$$

Lemma 1 (Conditional payoff dominance of unanimity)**.** $\alpha < \alpha^*(N,\beta)$ *if and only if, for every* $\mu \in (0,1]$,

$$E_\theta[V_H^{R1}(\theta,\mu,U)] > E_\theta[V_H^{R1}(\theta,\mu,M)].$$

Proof. See Appendix B.5. □

The condition $\alpha < \alpha^*(N,\beta)$ is pinned down by the no-uncertainty endpoint $\mu = 1$. At that belief, screening rents vanish: the hegemon is known to be the high type, so unanimity can dominate majority only if its bargaining advantage exceeds the cost of buying all votes. The appendix shows that this is the tightest case: once $D(1) > 0$, the conditional payoff advantage of unanimity, $D(\mu) \equiv E[V_H(\mu,U)] - E[V_H(\mu,M)]$, remains strictly positive for all beliefs $\mu \in (0,1]$. The intuition has two parts. When the hegemon proposes, unanimity depresses weak states' continuation values (because rejection leads to a game where the hegemon is pivotal), allowing the hegemon to buy agreement more cheaply. When a weak state proposes, unanimity forces the proposer to secure the hegemon's vote, generating the screening rent identified in Proposition 3. Both forces favor the hegemon.

Substantively, α measures the strength of the hegemon's bilateral outside option relative to multilateral cooperation. The threshold α^* identifies when the cost of buying unanimous agreement—which requires $N-1$ votes, versus $q-1$ under majority—outweighs the screening advantage. Because unanimity's vote-buying cost grows with N while majority's grows more slowly, α^* decreases with organization size: for $\beta = 0.9$, $\alpha^* \approx 0.47$ when $N = 5$ but falls to ≈ 0.03 when $N = 164$. The condition is therefore easy to satisfy in small institutions but demanding in large ones. For a WTO-sized organization, conditional dominance of unanimity requires that the hegemon's bilateral alternatives capture only a small fraction of the multilateral surplus. This does not eliminate the mechanism—the screening jump persists regardless of α—but it restricts the domain over which unanimity dominates at *every* belief. Section 8.2 discusses what happens outside this domain.

Remark 1 (Robustness beyond α^*)**.** *When $\alpha \geq \alpha^*$, the conditional advantage of unanimity fails only near $\mu = 1$. In Region III, the payoff difference decomposes as*

$$D(\mu) = \underbrace{D(1)}_{\text{without info}} + (1-\mu) \cdot \underbrace{\Gamma}_{\text{value of uncertainty}},$$

where $C_{buy} \equiv \beta(q-1)(1-\alpha)$ is the vote-buying cost under majority, $C_{out} \equiv N(N-1)\alpha$ is the aggregate outside-option cost, $D(1) = r[C_{buy} - C_{out}(1-\beta)]/N^2 \leq 0$ is the payoff comparison when the hegemon's type is known, and $\Gamma = (r-1)[C_{out} - C_{buy} + (N-1)\beta]/N^2 > 0$ is the marginal value of uncertainty for unanimity. The belief threshold above which majority dominates is $\bar{\mu} = 1 - |D(1)|/\Gamma$. Numerically, $\bar{\mu}$ is remarkably stable: for $N = 164$ and $\beta = 0.9$, increasing α from 0.05 to 0.49 (an 18-fold increase beyond α^*) only lowers $\bar{\mu}$ from 0.87 to 0.71 when $r = 1.5$. The mechanism fails only near certainty about the magnitude of the gains from cooperation—precisely where private information has least value (Figure
reffig:heatmap-alpha-mu in Section
refscope maps the full (α, μ) frontier).

An immediate corollary is that entry is always harder under unanimity ($E_U \subseteq E_M$): because the hegemon captures more surplus (Lemma 1), less is available for weak states, so $V_W(\mu, U) < V_W(\mu, M)$ for every $\mu \in (0, 1]$ (Appendix B.6). Without persuasion, the institutional comparison is therefore ambiguous: unanimity dominates when the institution forms under both rules, but majority dominates when only it can induce entry.

7.2 The main result

With Bayesian persuasion, the hegemon's payoff under each rule is $\Pi^*_H(R, p) = \alpha V_e(p) + \text{cav}\, v(p, R)$. Since the outside-option term is common to both rules, the institutional choice reduces to comparing $\text{cav}\, v(p, U)$ versus $\text{cav}\, v(p, M)$. Let $E_U \equiv \{\mu \in (0, 1] : V_W^{R1}(\mu, U) \geq c\}$ denote the set of beliefs at which weak states enter under unanimity. The central result is pointwise: at every belief at which unanimity can induce entry, the hegemon strictly prefers unanimity to majority.

Theorem 1 (Dominance of unanimity)**.** *Suppose $\alpha < \alpha^*(N, \beta)$. For every prior $p \in E_U$,*

$$\Pi_H^*(U, p) > \Pi_H^*(M, p).$$

That is, whenever entry is feasible under unanimity, the hegemon strictly prefers it.

Proof. See Appendix B.6. □

The theorem holds regardless of whether E_U is connected or disconnected, because it applies pointwise to every prior at which unanimity entry occurs. The screening mechanism (Proposition 3) gives the hegemon a higher conditional payoff, and majority's linearity (Proposition 1) means that concavification cannot close the gap. The only channel through which majority can dominate is the entry margin.

Corollary 1 (Global dominance when entry is everywhere feasible)**.** *If $E_U = (0, 1]$—entry costs are low enough that the institution forms under unanimity at all beliefs—then $\Pi_H^*(U, p) > \Pi_H^*(M, p)$ for every $p \in (0, 1]$.*

When $E_U \neq (0, 1]$, the comparison at beliefs outside the unanimity entry set depends on how persuasion shifts posteriors into E_U. The entry set E_U may be disconnected: V_W^{R1} can have a downward jump at μ_s^{R1} (the mirror of V_H's upward jump) that creates a gap around the screening cutoff (Appendix A.7). The following result holds regardless of whether E_U is connected.

Theorem 2 (Single-crossing institutional comparison)**.** *Suppose $\alpha < \alpha^*(N, \beta)$. The set of priors for which unanimity strictly dominates majority,*

$$\{p \in (0, 1] : \Pi_H^*(U, p) > \Pi_H^*(M, p)\},$$

is an upper interval. That is, there exists $p^* \in [0, 1]$ such that majority weakly dominates for $p < p^*$ and unanimity strictly dominates for $p > p^*$. Any advantage of majority arises exclusively through the entry margin. This result does not require E_U to be connected.

Proof. See Appendix B.8. The proof uses a supporting lemma (Appendix B.7) showing that $V_W^{R1}(1, U)$ is the global maximum of $V_W^{R1}(\mu, U)$, which ensures $1 \in E_U$ whenever $E_U \neq \emptyset$. □

Theorem 2 says that the institutional ranking never oscillates. At sufficiently high priors, unanimity dominates because the screening mechanism gives the hegemon a conditional payoff advantage. At pessimistic priors, majority can dominate—but only through the entry margin, never through a higher conditional payoff. The transition occurs at most once. The result holds regardless of whether the unanimity entry set E_U is connected or disconnected.

Example 2 (Computing p^*: the full institutional comparison)**.** *Continuing with the parameters from Example 1 (*$N = 5$*,* $r = 1.5$*,* $\alpha = 0.3$*,* $\beta = 0.9$*), suppose entry costs* $c = 0.14$*. Under majority, the institution forms at essentially all beliefs (*$\tau(M) \approx 0$*). Under unanimity, the screening mechanism reduces weak states' surplus, so the institution forms only for* $\mu > \tau(U) \approx 0.33$*. Bayesian persuasion partially closes this entry gap: the hegemon designs signals that push posteriors above* $\tau(U)$*, inducing participation under unanimity at priors below* 0.33*.*

The threshold is $p^* \approx 0.24$*. Below it, majority dominates because only it can induce entry—unanimity's concavified payoff cannot overcome the participation disadvantage. Above it, unanimity strictly dominates: at* $p = 0.30$*, the hegemon's payoff under unanimity exceeds majority by* 22%*; at* $p = 0.50$*, by* 25%*. The advantage of majority at pessimistic priors is entirely through the entry channel: conditional on participation, unanimity always gives the hegemon more (Lemma 1).*

The transition is sharp and monotone, as Theorem 2 guarantees. The set of priors favoring unanimity is the upper interval $(0.24, 1]$*: once the prospects of cooperation are promising enough that unanimity can induce entry (directly or through persuasion), the screening advantage is decisive.*

7.3 Numerical characterization

Figure 5 shows that the hegemon prefers unanimity across a broad range of parameters when the prior is low ($p = 0.05$). In the blue regions, unanimity dominates globally (Corollary 1 or because p^* lies below 0.05). The unanimity-dominance region is larger when the hegemon's private information is more consequential (higher r) and when bargaining is more patient (higher β). In the red regions, Theorem 2 applies: the threshold p^* lies above 0.05, so majority dominates at this pessimistic prior, but unanimity would dominate at higher priors.

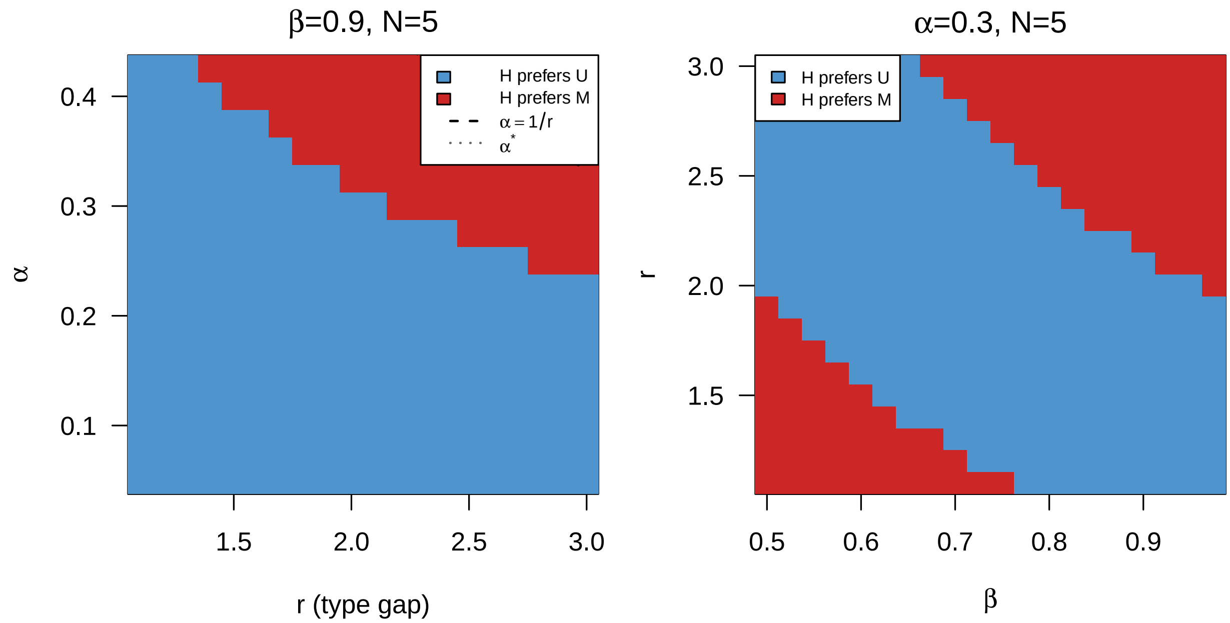

Figure 5: Parameter regions where the hegemon prefers unanimity (blue) vs majority (red) at prior $p = 0.05$, restricted to $\alpha < \alpha^*(N, \beta)$ (the domain where Lemma 1 applies). Left: r vs α plane with $\beta = 0.9$, $N = 5$, $c = 0.1$. Right: β vs r plane with $\alpha = 0.3$, $N = 5$, $c = 0.1$.

8 Discussion

8.1 Illustration: The GATT/WTO

The WTO illustrates the interaction between consensus and informational asymmetry. The multilateral trading system operates by consensus, even though formal voting provisions exist (Steinberg 2002). The United States and other major trading powers could plausibly press for majority rule, yet consensus persists.

The model's informational asymmetry assumption finds support in the trade context. Evaluating the effects of tariff schedules, anticipating distributive consequences, and assessing long-term implications of rule changes require sustained investment in analytical infrastructure. Major trading powers maintain large permanent delegations with deep sectoral expertise; many developing country delegations operate with far smaller teams, particularly during complex technical phases of negotiations. This capacity asymmetry means that weaker states often face genuine uncertainty about the value of proposed agreements.

In practice, major trading powers exercise informal agenda influence through the Green Room process, control over drafting, and technical expertise. The model's baseline uses symmetric proposal rights to isolate the voting-rule comparison, but the qualitative mechanism—screening under unanimity, no screening under majority—does not depend on equal recognition probabilities. Monopoly agenda control would eliminate participation (as argued in Section 8.2), so the relevant question is whether moderate informal influence coexists with the screening channel. The persistence of consensus is consistent with the coexistence of informal influence and formal equality.

The model's mechanism explains a pattern that existing accounts describe but do not fully explain: powerful states in the WTO appear to benefit from consensus despite lacking formal agenda control. The mechanism is not merely that informal practices survive consensus, but that consensus creates the strategic environment—pivotal inclusion under uncertainty—in which informational power becomes most valuable. The model does not claim that the GATT/WTO was designed for this reason, or that all aspects of WTO politics reduce to this mechanism. It shows that consensus and informational asymmetry interact in a way that can benefit powerful states, offering one explanation for why hegemonic preference for consensus is not paradoxical.

The informational mechanism generates several observable implications that distinguish it from alternative accounts of consensus. The hegemon's advantage from consensus should be largest in negotiations where informational asymmetry is most consequential for the value of cooperation. In the WTO, this suggests that consensus should matter most in complex regulatory areas—services, intellectual property, investment rules—where evaluating proposals requires deep technical expertise, and least in areas where distributive consequences are transparent, such as linear tariff cuts. Legitimacy accounts, which predict that consensus matters uniformly across issue areas, do not generate this cross-issue variation.[8]

The mechanism also predicts that the hegemon's preference for consensus should weaken as its bilateral alternatives improve. The proliferation of preferential trade agreements strengthens the hegemon's outside option, narrowing the screening advantage that makes unanimity valuable (Bhagwati 2008). Conversely, consensus should emerge in mature regimes with established cooperation value, not in new or untested institutional settings: when the primary challenge is inducing participation,

[8] Appendix C shows that with $K > 2$ types, unanimity generates $K - 1$ screening boundaries while majority remains linear. More issue dimensions increase the number of screening regions but also make the screening problem harder for the hegemon, suggesting that the mechanism is strongest in focused negotiations.

the easier entry conditions under majority outweigh the screening advantages of unanimity. The screening advantage is amplified by bargaining patience, suggesting that consensus should be most valuable in negotiations with long time horizons. Self-enforcement accounts, which predict that consensus matters most when enforcement is weak regardless of information, do not generate this dependence on informational asymmetry or bargaining patience.

8.2 Scope

Why symmetric proposal rights? Under either voting rule, if the hegemon were the exclusive proposer and weak states had no outside option from bargaining ($d_W = 0$), weak states' expected bargaining payoff would be zero. No weak state would pay an entry cost $c > 0$, and the institution would never form. Monopoly proposal power is self-defeating when participation is voluntary. The relevant institutional choice is therefore between voting rules, holding proposal rights symmetric.

Why not compare consensus with hegemonic agenda control? Because agenda control combined with any participation constraint eliminates the institution itself. The puzzle is not why a hegemon does not impose dictatorship, but why it would choose a rule that makes its own approval necessary when majority would allow others to bypass it.

When consensus favors the hegemon. The mechanism requires that: (i) informational asymmetry is relevant (H knows θ, W does not); (ii) the hegemon has a valuable outside option ($\alpha > 0$); (iii) entry is costly and belief-dependent; (iv) the voting rule makes H pivotal.

The model assumes the hegemon can commit to an information structure before observing θ. This is a strong assumption that warrants discussion. Three considerations support its plausibility in the IO context. First, *reputation*: in repeated interactions, a hegemon that deviates from an announced disclosure policy suffers credibility costs in future negotiations, which can sustain commitment in equilibrium. Second, *institutional transparency rules*: organizations like the WTO have formal provisions for notifications, trade policy reviews, and dispute settlement that constrain and channel information disclosure, serving as partial commitment devices. Third, *upper bound interpretation*: the Bayesian persuasion payoff represents the maximum value of the hegemon's informational advantage. Even if full commitment is unavailable, the qualitative institutional comparison—unanimity creates screening opportunities that majority does not—may survive under weaker information structures, such as evidence games (Glazer and Rubinstein 2004) or verifiable disclosure (Milgrom 1981).

When majority dominates. The biconditional in Lemma 1 implies that $\alpha < \alpha^*$ is the largest parametric region where majority can never dominate through a higher conditional-on-entry payoff. Any advantage of majority must come through the entry margin: $\tau(U) > \tau(M)$. This is more likely when entry costs are high (creating a large gap where the institution forms only under majority), when α approaches α^* (weakening the conditional dominance), or when information is approximately public. For $\alpha \geq \alpha^*$, majority can also dominate through conditional bargaining payoffs near $\mu = 1$, because unanimity's cost of buying $N - 1$ votes outweighs the screening advantage.

The threshold α^* decreases with N because unanimity requires buying $N-1$ votes, while majority requires only $q - 1$. For large organizations, α^* becomes small: the conditional dominance result applies only when the hegemon's bilateral alternative is modest relative to multilateral cooperation. This does not eliminate the mechanism: even when Lemma 1 fails globally, the screening jump at μ_s^{R1} persists, and Bayesian persuasion can still exploit it at low priors. The condition restricts when unanimity dominates *everywhere*, not when it dominates *somewhere*. Figure 6 illustrates: comparing the $N = 30$ and $N = 164$ panels, α^* (dashed line) shifts left, but the blue region—where unanimity dominates—remains extensive.

The relationship between α^* and bilateral alternatives has a substantive implication. The proliferation of preferential trade agreements can be understood, in the model's terms, as an increase in α: bilateral and regional deals strengthen the hegemon's outside option relative to multilateral cooperation (Bhagwati 2008). As α rises past α^*, the hegemon's institutional preference shifts away from consensus—not because multilateral cooperation is less valuable, but because the hegemon can capture a larger share of that value bilaterally.

The biconditional in Lemma 1 implies that this boundary is sharp. At $\alpha = \alpha^*$, the dominance difference $D(\mu)$ equals zero exactly at $\mu = 1$; for $\alpha > \alpha^*$, majority strictly dominates near $\mu = 1$ (where the vote-buying cost is decisive), while unanimity can still dominate at lower beliefs. The paper's main results are stated for $\alpha < \alpha^*$, which ensures a clean separation between the conditional and entry channels.

Figure 6 maps the conditional dominance region in the (α, μ) plane. For each parameter combination, the blue region indicates beliefs at which unanimity gives the hegemon a higher continuation payoff; the red region indicates beliefs at which majority dominates. The solid curve traces the frontier $\bar{\mu}(\alpha)$ from Remark 1. Two patterns stand out. First, the frontier is remarkably flat: increasing α well beyond α^* shifts $\bar{\mu}$ only modestly downward, confirming that the screening advantage is robust across most of the belief space. Second, the red region is always concentrated near $\mu = 1$—near certainty about the magnitude of the gains from cooperation—where informational asymmetry is irrelevant and the cost of buying unanimous agreement dominates.

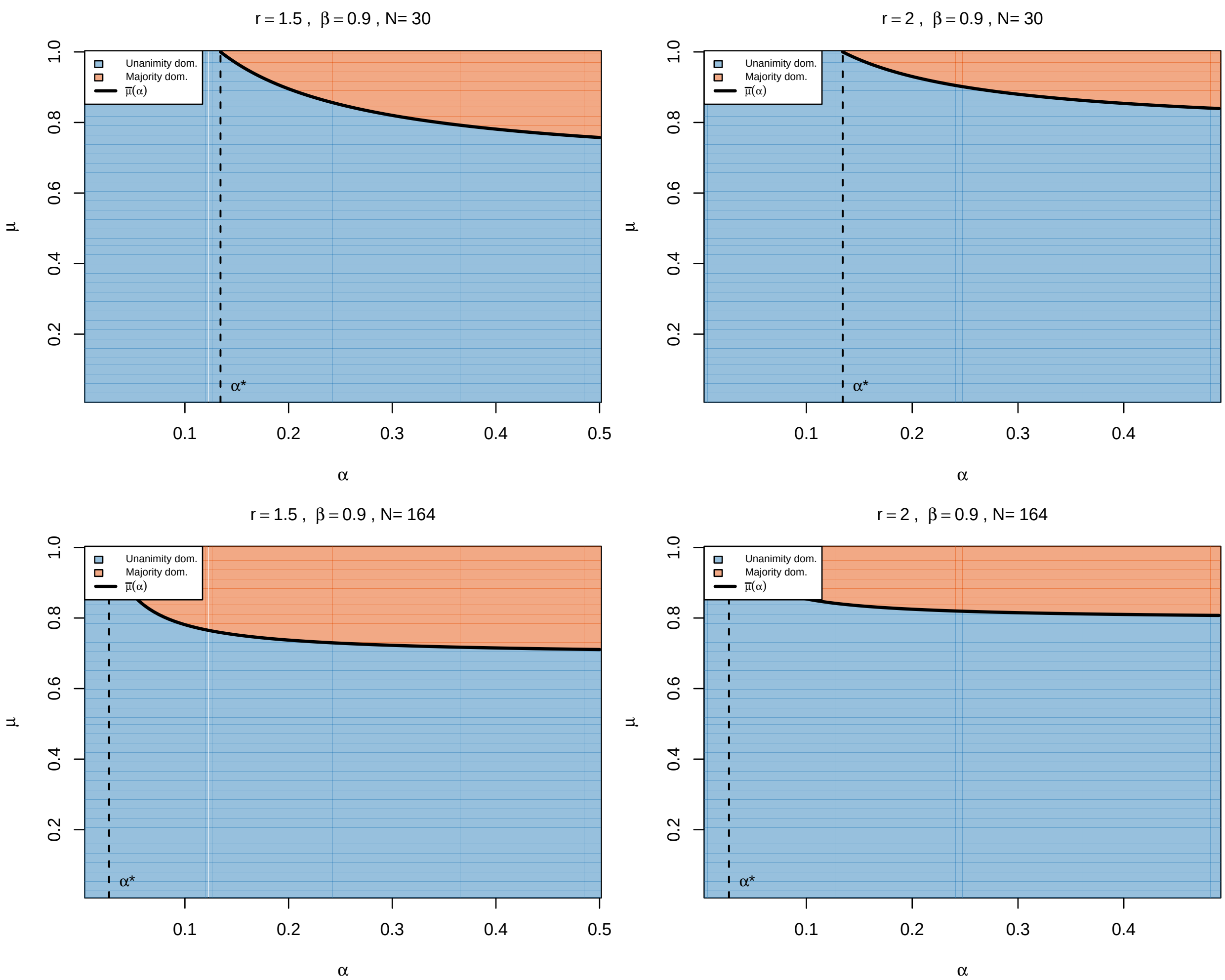

Figure 6: Conditional payoff dominance in the (α, μ) plane. Blue: unanimity dominates ($D(\mu) > 0$). Red: majority dominates ($D(\mu) < 0$). Solid curve: belief threshold $\bar{\mu}(\alpha)$ above which majority dominates (Remark 1). Dashed line: α^*. Left column: $r = 1.5$. Right column: $r = 2$. Top row: $N = 30$. Bottom row: $N = 164$. All panels: $\beta = 0.9$.

Alternative explanations. The informational mechanism is complementary to, not a substitute for, other rationalist explanations of consensus. Legitimacy and compliance accounts emphasize that consent enhances durability of agreements, but address why agreements are honored rather than why a hegemon benefits from the bargaining structure. Self-enforcing agreement models (Maggi and Morelli 2006) show that unanimity can be optimal when enforcement is weak, but apply primarily to symmetric states and collective action problems rather than to asymmetric players with private information. Issue linkage accounts explain why broad agendas favor consensus but not why a hegemon with superior information would prefer unanimity in a single-issue environment. This paper isolates a distributive-informational mechanism within the rationalist framework common to these accounts, in the spirit of Fearon (1995).

Broader implications. Institutions do not merely reflect power; they select which form of power becomes politically productive. Majority rule rewards coalition arithmetic; unanimity rewards informational influence. This reframing implies a dynamic: as weaker states invest in independent analytical capacity, the informational asymmetry narrows, the returns to persuasion decline, and the hegemon's institutional incentive to prefer consensus should weaken. These dynamic implications are suggestive extrapolations from the static mechanism, not formal predictions of the model. A proper treatment would require a repeated game with endogenous information acquisition, which is beyond the scope of this paper. The value of consensus to a hegemon should covary with the distribution of technical expertise across the membership.

9 Conclusion

This paper has identified conditions under which consensus is a rational choice for a hegemon. The key is to distinguish between two modes of exercising power under different voting rules. Majority rule with symmetric proposals lets weak states bypass the hegemon, eliminating the screening problem. Unanimity forces weak states to include the hegemon under uncertainty, creating a screening cutoff that gives the hegemon a conditional payoff advantage at every level of uncertainty. Bayesian persuasion extends this advantage to priors where the institution would not otherwise form. The institutional comparison reduces to a single threshold: above it, unanimity dominates; below it, majority can dominate through easier entry. The ranking never oscillates.

The argument does not imply that all consensus institutions are designed for this reason, or that informal power disappears under consensus. It shows something more specific: under identifiable conditions on informational asymmetry and the prospects of cooperation, unanimity becomes the preferred institutional arrangement for a hegemon. Consensus may be less a concession by power than a way of exercising power through a different channel—the channel of information.

The mechanism depends on a discrete type space. With $K > 2$ types, Appendix C shows that unanimity generates $K - 1$ screening boundaries while majority remains linear—the non-convexity structure becomes richer, not weaker. With a continuous type space, however, the offer schedule would generically be smooth, eliminating the discrete jumps that Bayesian persuasion exploits. Discrete types are substantively natural in this setting—states evaluate cooperation in terms of qualitatively distinct regimes rather than marginal variations—but whether informational power retains strategic value under continuous types is an open question.

Several extensions remain. First, if weaker states can invest in independent analytical capacity, informational power erodes endogenously over time. Second, heterogeneous outside options across weak states (middle powers) would enrich the screening dynamics. Third, repeated play within an established consensus institution could generate learning that narrows the informational gap. These directions point toward a richer theory of how informational power rises and declines within institutional frameworks.

Appendix A: Bargaining Derivations

Notation summary

Symbol	Meaning	Defined in
N	Number of players (1 hegemon $+N-1$ weak states)	Def. 1
$\theta \in \{0,1\}$	State of the world	Def. 1
p	Prior Pr$(\theta = 1)$	Def. 1
μ	Posterior belief Pr$(\theta = 1 \mid s)$	Sec. 3
$V(\theta)$	Value of cooperation: $V(0) = 1, V(1) = r$	Def. 1
$r > 1$	High-type value	Def. 1
$\alpha \in (0, 1/r)$	Outside-option share: $d_H = \alpha V(\theta)$	Def. 1
$\beta \in (0,1)$	Common discount factor	Def. 1
$c > 0$	Entry cost per weak state	Def. 1
$V_e(\mu)$	Expected cooperation value $1 + \mu(r-1)$	Sec. 3
x	Shorthand $(N-1)\alpha r$	Sec. 3
q	Majority quota $\lfloor N/2 \rfloor + 1$	Def. 1
$R \in \{M, U\}$	Voting rule (majority / unanimity)	Def. 1
μ_s^{R2}	R2 screening cutoff	App. A.2
μ_s^{R1}	R1 screening cutoff	Prop. 2
ϕ	Auxiliary for R1 cutoff: $[rN - \beta(N-1+r)]/[\beta(r-1)]$	Prop. 2
$\tau(R)$	Entry threshold under rule R	Sec. 6
$v(\mu, R)$	Hegemon's net gain from institution: $E[V_H^{R1}] - \alpha V_e(\mu)$ if entry, 0 otherwise	Sec. 6
$\bar{\alpha}$	Regime threshold for α-independent R1 cutoff	Prop. 2
α^*	Iff threshold for conditional dominance (Lemma 1)	Eq. (4)
$C_{\text{buy}}, C_{\text{out}}$	Vote-buying cost, outside-option cost (proof)	App. B.5
λ_M	Majority payoff coefficient	Thm. 2
S_U, S_M	Max-slope of value function under U, M	Thm. 2
p^*	Threshold prior for institutional comparison	Thm. 2

A.1 Majority: R2 and R1

Under majority, weak proposers form a winning coalition from other weak states, excluding H. The hegemon captures $\alpha V(\theta)$ from its bilateral alternative. H's payoff is type-specific and does not depend on W's beliefs.

R2 continuation values:

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1 + (N-1)\alpha]}{N} \tag{5}$$

$$V_W^{R2}(\mu, M) = \frac{(1-\alpha)V_e(\mu)}{N} \tag{6}$$

R1 values follow by backward induction with the same structure (no screening, linear).

A.2 Unanimity: R2

When H proposes: offers 0 to all W's, keeps $V(\theta)$.

When W proposes: the screening subgame has cutoff $\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$. Below it, W plays aggressive ($y_H = \alpha$); above, conservative ($y_H = \alpha r$).

The hegemon's R2 continuation values under unanimity are:

$$V_H^{R2}(\theta = 1, \mu) = \frac{r[1 + (N-1)\alpha]}{N} \quad \text{(constant in } \mu\text{)}, \tag{7}$$

$$V_H^{R2}(\theta = 0, \mu < \mu_s^{R2}) = \frac{1 + (N-1)\alpha}{N}, \tag{8}$$

$$V_H^{R2}(\theta = 0, \mu > \mu_s^{R2}) = \frac{1 + (N-1)\alpha r}{N} \quad \text{(overpaid).} \tag{9}$$

Weak states' R2 continuation values under unanimity:

$$V_W^{R2}(\mu < \mu_s^{R2}) = \frac{(1-\mu)(1-\alpha)}{N} \tag{10}$$

$$V_W^{R2}(\mu > \mu_s^{R2}) = \frac{V_e(\mu) - \alpha r}{N} \tag{11}$$

Continuity at μ_s^{R2} is verified by direct substitution.

A.3 Unanimity: R1 screening derivation

The proposing W compares:

$$F_1^{con}(\mu) = V_e(\mu) - \frac{\beta(r+x)}{N} - \omega(\mu) \tag{12}$$

$$F_1^{agg}(\mu) = (1-\mu)\left[1 - \frac{\beta(1+x)}{N} - \omega(\mu)\right] + \frac{\mu\beta r(1-\alpha)}{N} \tag{13}$$

where $\omega(\mu) = (N-2)\beta V_W^{R2}(\mu)$.

The difference $\Delta_1(\mu) = F_1^{agg} - F_1^{con}$ is piecewise quadratic. On the high branch ($\mu > \mu_s^{R2}$), the α-dependent terms cancel, yielding the quadratic:

$$(N-2)\beta(r-1)\mu^2 + [\beta(N-1+r) - rN]\mu + \beta(r-1) = 0.$$

Dividing by $\beta(r-1)$:

$$(N-2)\mu^2 - \phi\mu + 1 = 0, \qquad \phi = \frac{rN - \beta(N-1+r)}{\beta(r-1)}.$$

The smaller root is μ_s^{R1} (equation (1)).

A.4 Jump magnitude

At the R1 cutoff, type $\theta = 0$ receives $\beta(1+x)/N$ (aggressive) vs $\beta(r+x)/N$ (conservative) when W proposes. The difference $\beta(r-1)/N$, weighted by $(N-1)/N$ (prob W proposes) and $(1-\mu_s^{R1})$ (prob $\theta = 0$), gives the jump in equation (3).

A.5 Case selection

The R1 screening cutoff lies on the high R2 branch ($\mu_s^{R1} > \mu_s^{R2}$, α-independent) when $\Delta_1(\mu_s^{R2}) > 0$. On the high branch, Δ_1 is a convex quadratic with $\Delta_1(0) > 0$ and $\Delta_1(1) = rN(\beta-1) < 0$, so $\Delta_1(\mu) > 0$ for all $\mu < \mu_s^{R1}$. Hence $\Delta_1(\mu_s^{R2}) > 0$ if and only if $\mu_s^{R2} < \mu_s^{R1}$. Substituting

$\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$ into $\mu_s^{R2} < \mu_s^{R1}$ and solving for α:

$$\frac{\alpha(r-1)}{r-\alpha} < \mu_s^{R1} \quad \Longleftrightarrow \quad \alpha < \frac{\mu_s^{R1} \cdot r}{r-1+\mu_s^{R1}} = \bar{\alpha}(r,\beta,N),$$

which defines the threshold in equation (2).

When $\alpha \geq \bar{\alpha}$, the R1 cutoff falls on the low R2 branch, where $V_W^{R2}(\mu) = (1-\mu)(1-\alpha)/N$ and Δ_1 depends on α through $\omega(\mu)$. The cutoff still exists and is unique: $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ hold regardless of branch, so a root exists by the intermediate value theorem; since Δ_1 is quadratic on $(0, \mu_s^{R2})$ (at most two roots) with opposite-sign endpoints, there is exactly one root. On the high branch, $\Delta_1 \leq 0$, so no additional roots exist. The jump at μ_s^{R1} remains positive. The proof of Lemma~1 (Appendix~B.5) covers both regimes explicitly: the payoff decomposition $D = D_{\text{base}} + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2} + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}$ is additive regardless of the relative ordering of μ_s^{R1} and μ_s^{R2}, and the endpoint argument establishes $D(\mu) > 0$ on all branches in both cases.

A.6 Budget checks

Under majority, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds exactly: $V_H[1+(N-1)\alpha]/N + (N-1)(1-\alpha)V_e(\mu)/N = V_e(\mu) \cdot [1+(N-1)\alpha + (N-1)(1-\alpha)]/N = V_e(\mu)$. Under unanimity, the budget identity $E[V_H] + (N-1)V_W = V_e(\mu)$ holds on the conservative R1 branch, where all deals pass in R1. On the aggressive R1 branch, surplus destruction from discounting when $\theta = 1$ rejects in R1 and the game proceeds to R2 yields $E[V_H] + (N-1)V_W = V_e(\mu) - \frac{(N-1)\mu r(1-\beta)}{N}$, so the identity holds with inequality: $E[V_H] + (N-1)V_W \leq V_e(\mu)$.

A.7 Entry thresholds

Under majority, $V_W^{R1}(\mu, M) = \kappa_M V_e(\mu)$ where $\kappa_M = (1-\alpha)[N(N-1)+\beta(q-1)]/(N^2(N-1))$. The entry threshold is $\tau(M) = \max\{0,\ (c/\kappa_M - 1)/(r-1)\}$.

Under unanimity, V_W^{R1} is piecewise. On the conservative R1 branch ($\mu > \mu_s^{R1}$, which also implies $\mu > \mu_s^{R2}$ when $\alpha < \bar{\alpha}$):

$$V_W^{R1}(\mu, U) = \frac{(N+\beta)\,V_e(\mu) - \beta r(1+N\alpha)}{N^2}.$$

This is affine in μ. Setting $V_W^{R1} = c$ and solving:

$$\tau_U^{con} = \frac{cN^2 - [N + \beta - \beta r(1 + N\alpha)]}{(N + \beta)(r - 1)}. \tag{14}$$

On the aggressive R1 branch ($\mu < \mu_s^{R1}$), the proposer surplus F_1^{agg} is quadratic in μ (the product of $(1-\mu)$ with the affine terms in $\omega(\mu)$ generates a second-order term). However, the full weak-state expected payoff V_W^{R1}—which averages over proposer and non-proposer roles—simplifies to a piecewise affine expression on each R2 sub-branch, because the quadratic terms in the proposer and non-proposer components cancel (see the proof of Lemma 2 in Appendix B.7). The entry threshold on each sub-branch therefore solves a linear equation.[9]

The numerical results in the paper compute $\tau(U)$ by pointwise evaluation of V_W^{R1}.

Appendix B: Proofs

B.1 Proof of Proposition 1 (Majority produces no screening)

Under majority rule, W's proposal passes with $q - 1$ votes from other weak states. H's vote is not required. W excludes H from the winning coalition; H captures $\alpha V(\theta)$ from its bilateral alternative regardless of μ. There is no screening and no jump. All continuation values are affine in $V_e(\mu)$, hence affine in μ. $\square$

B.2 Proof of Proposition 2 (R1 cutoff existence and uniqueness)

See Appendix A.3 for the derivation. *Existence:* $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(1) = rN(\beta - 1) < 0$; by continuity, a root exists in $(0, 1)$.

Uniqueness and closed form under $\alpha < \bar{\alpha}$: When $\alpha < \bar{\alpha}(r, \beta, N)$, the root lies on the high R2 branch ($\mu > \mu_s^{R2}$), where Δ_1 is a convex quadratic in μ with coefficients independent of α (Appendix A.3). Since $\Delta_1(\mu_s^{R2}) > 0$ (Appendix A.5) and $\Delta_1(1) < 0$, there is exactly one root

[9] Because V_W^{R1} has a downward jump at μ_s^{R1} (opposite to V_H's upward jump), the entry set $\{\mu : V_W^{R1} \geq c\}$ can be disconnected for intermediate values of c: entry may occur both below μ_s^{R1} (aggressive regime) and above some $\mu_2 > \mu_s^{R1}$ (conservative regime), with a gap around the cutoff. The affine formula (14) gives the entry threshold in the conservative regime and is the relevant quantity when $c > \max_{\mu \leq \mu_s^{R1}} V_W^{R1}(\mu, \text{agg})$.

in $(\mu_s^{R2}, 1)$, given by equation (1). Below μ_s^{R2}, Δ_1 is a concave quadratic in μ: the coefficient of μ^2 is $-(N-2)\beta(1-\alpha)/N < 0$, arising from the product $(1-\mu)\cdot\omega(\mu)$ in F_1^{agg} where $\omega(\mu) = (N-2)\beta(1-\mu)(1-\alpha)/N$ is linear in μ. Since $\Delta_1(0) = \beta(r-1)/N > 0$ and $\Delta_1(\mu_s^{R2}) > 0$, and a concave function that is positive at both endpoints of an interval is positive throughout, $\Delta_1(\mu) > 0$ for all $\mu \in [0, \mu_s^{R2}]$. No additional roots exist.

Uniqueness under $\alpha \geq \bar{\alpha}$: The root falls on the low R2 branch, where Δ_1 is a quadratic in μ that depends on α. On this branch, $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$ (since $\mu_s^{R2} \geq \mu_s^{R1}$ on the high branch implies $\Delta_1^{\text{high}}(\mu_s^{R2}) \leq 0$, and continuity gives $\Delta_1(\mu_s^{R2}) \leq 0$). Since Δ_1 is quadratic on $(0, \mu_s^{R2})$, it has at most two roots; combined with $\Delta_1(0) > 0$ and $\Delta_1(\mu_s^{R2}) \leq 0$, there is exactly one root in $(0, \mu_s^{R2}]$. On the high branch, $\Delta_1 \leq 0$ for $\mu \in [\mu_s^{R2}, 1]$, so no additional roots exist there. □

B.3 Proof of Proposition 3 (Jump)

Direct computation from the difference in H's R1 payoff between the conservative and aggressive branches. See Appendix A.4. □

B.4 Proof of Proposition 4 (Additional non-convexity)

Under majority, $v(\mu, M)$ is affine for $\mu \geq \tau(M)$, hence concave—the concave envelope coincides with v above the threshold. Under unanimity, the jump at μ_s^{R1} means $v(\mu, U)$ is not concave on $[\tau(U), 1]$: $\lim_{\mu\uparrow\mu_s^{R1}} v(\mu, U) < \lim_{\mu\downarrow\mu_s^{R1}} v(\mu, U)$. The concave envelope must therefore lie strictly above $v(\mu, U)$ in a neighborhood of μ_s^{R1}. Since $v(\mu, U)$ is affine on each branch (the expected payoff is linear in μ conditional on the weak proposer's strategy), the upward jump is the sole source of non-concavity. □

B.5a Derivation of the payoff decomposition

This appendix derives the decomposition $D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu)$ from the primitive payoff formulas in Appendices A.1–A.3.

Step 0: Primitive payoff components. The first task is to write down the hegemon's R1 payoff from its primitive components. Because proposals are random, the payoff splits naturally by proposer identity, and this separation will prove essential: each component turns out to depend on a different screening regime, which is ultimately why the two corrections are additive.

The hegemon's expected R1 payoff $E[V_H^{R1}(\mu, U)]$ has two components, corresponding to the identity of the R1 proposer.

H proposes (probability $1/N$). The hegemon offers each weak state $\beta V_W^{R2}(\mu)$ (its discounted R2 continuation) and keeps the remainder. The expected payoff across types is:

$$\Pi_H^{\text{prop}}(\mu) = \frac{V_e(\mu) - (N-1)\beta V_W^{R2}(\mu)}{N}. \tag{15}$$

W proposes (probability $(N-1)/N$). The proposing weak state makes an offer to H that depends on the R1 screening regime. Under an aggressive R1 offer ($\mu < \mu_s^{R1}$), only $\theta = 0$ accepts; $\theta = 1$ rejects and the game proceeds to R2 with posterior $\mu' = 1$. The R1 offer to H is $\beta V_H^{R2}(\theta{=}0, \mu'{=}1)$, and $\theta = 1$'s payoff from rejection is $\beta V_H^{R2}(\theta{=}1, \mu'{=}1)$. Since $\mu' = 1 > \mu_s^{R2}$, both R2 values are evaluated in the conservative R2 regime (Appendix A.2, equations (7)–(9)):

$$\beta V_H^{R2}(\theta{=}0, \mu'{=}1) = \frac{\beta(1+x)}{N}, \tag{16}$$

$$\beta V_H^{R2}(\theta{=}1, \mu'{=}1) = \frac{\beta(r+x)}{N}. \tag{17}$$

Under a conservative R1 offer ($\mu > \mu_s^{R1}$), both types accept. The offer is set at $\beta V_H^{R2}(\theta{=}1, \mu'{=}1) = \beta(r+x)/N$ to satisfy the high type's participation constraint. Hence the expected payoff from W's proposal is:

$$\Pi_H^W(\mu) = \frac{N-1}{N} \begin{cases} \mu \dfrac{\beta(r+x)}{N} + (1-\mu)\dfrac{\beta(1+x)}{N} & \text{if } \mu < \mu_s^{R1} \text{ (aggressive R1)}, \\ \dfrac{\beta(r+x)}{N} & \text{if } \mu > \mu_s^{R1} \text{ (conservative R1)}. \end{cases} \tag{18}$$

The total hegemon payoff is $E[V_H^{R1}(\mu, U)] = \Pi_H^{\text{prop}}(\mu) + \Pi_H^W(\mu)$.

Two features are important. First, the R1 offers to H in (16)–(17) are evaluated at the off-path posterior $\mu' = 1$ and therefore do not depend on $V_W^{R2}(\mu)$. Second, H's proposal payoff (15) depends on $V_W^{R2}(\mu)$ but is the same regardless of the R1 regime. These two observations establish that the R1 and R2 corrections are independent.

Step 1: Benchmark payoff (aggressive R1, conservative R2). I now compute the hegemon's total payoff in a benchmark regime—aggressive R1 screening and conservative R2 screening—which serves as the reference point against which the two corrections will be measured. The choice of benchmark is not arbitrary: under this regime, both components of the payoff take their simplest form, and the corrections for each alternative regime can then be isolated one at a time.

Under aggressive R1 and conservative R2 ($V_W^{R2} = [V_e(\mu) - \alpha r]/N$), the two components are:

$$\Pi_H^{\text{prop}}(\mu)\big|_{\text{con-}R2} = \frac{V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]/N}{N} = \frac{N\,V_e(\mu) - (N-1)\beta[V_e(\mu) - \alpha r]}{N^2}, \tag{19}$$

$$\Pi_H^W(\mu)\big|_{\text{agg-}R1} = \frac{(N-1)\beta}{N} \cdot \frac{V_e(\mu) + x}{N} = \frac{(N-1)\beta[V_e(\mu) + x]}{N^2}. \tag{20}$$

Summing:

$$\begin{aligned} E_{\text{bench}}(\mu) &= \frac{N\,V_e(\mu) - (N-1)\beta V_e(\mu) + (N-1)\beta\alpha r + (N-1)\beta V_e(\mu) + (N-1)\beta x}{N^2} \\ &= \frac{N\,V_e(\mu) + (N-1)\beta(\alpha r + x)}{N^2} = \frac{N\,V_e(\mu) + (N-1)\beta N\alpha r}{N^2}, \end{aligned} \tag{21}$$

where the last equality uses $\alpha r + x = \alpha r + (N-1)\alpha r = N\alpha r$.

Step 2: R1 correction δ_{R1}. The next step is to compute the payoff change when the R1 regime switches from aggressive to conservative. Because this switch affects only the R1 offers that W makes to H—and not H's own proposal, which depends on V_W^{R2}—the correction will appear exclusively in Π_H^W, confirming that it can be treated independently of the R2 regime.

When the R1 regime switches from aggressive to conservative ($\mu > \mu_s^{R1}$), Π_H^{prop} is unchanged (it depends only on V_W^{R2}, not on the R1 regime). The change is entirely in Π_H^W. Under aggressive R1:

$$\Pi_H^W\big|_{\text{agg}} = \frac{(N-1)\beta}{N^2}[\mu(r+x) + (1-\mu)(1+x)] = \frac{(N-1)\beta[V_e(\mu) + x]}{N^2}.$$

Under conservative R1:

$$\Pi_H^W\big|_{\text{con}} = \frac{(N-1)\beta(r+x)}{N^2}.$$

The correction is:

$$\delta_{R1}(\mu) \equiv \Pi_H^W\big|_{\text{con}} - \Pi_H^W\big|_{\text{agg}} = \frac{(N-1)\beta}{N^2}[(r+x) - V_e(\mu) - x] = \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \tag{22}$$

This is non-negative for all $\mu \leq 1$, with equality only at $\mu = 1$. The correction reflects the overpayment that the low type receives under conservative R1 offers.

Step 3: R2 correction δ_{R2}. I now derive the payoff change when the R2 regime switches from conservative to aggressive. By an argument symmetric to Step 2, this switch affects only H's proposal payoff Π_H^{prop}—the R1 offers to H are pinned at the off-path posterior $\mu' = 1$, which always lies in the conservative R2 regime regardless of the on-path belief μ.

When the R2 regime switches from conservative to aggressive ($\mu < \mu_s^{R2}$), Π_H^W is unchanged (the R1 offers to H are evaluated at $\mu' = 1$, which is always in the conservative R2 regime). The change is entirely in Π_H^{prop}. Substituting $V_W^{R2,\text{agg}}(\mu) = (1-\mu)(1-\alpha)/N$ (Appendix A.2) in place of $V_W^{R2,\text{con}}(\mu) = [V_e(\mu) - \alpha r]/N$:

$$\begin{aligned}
\delta_{R2}(\mu) &\equiv \Pi_H^{\text{prop}}\big|_{\text{agg-}R2} - \Pi_H^{\text{prop}}\big|_{\text{con-}R2} \\
&= \frac{-(N-1)\beta}{N^2}\Big[(1-\mu)(1-\alpha) - V_e(\mu) + \alpha r\Big] \\
&= \frac{(N-1)\beta}{N^2}\Big[V_e(\mu) - \alpha r - (1-\mu)(1-\alpha)\Big].
\end{aligned} \tag{23}$$

Expanding the bracket:

$$V_e(\mu) - \alpha r - (1-\mu)(1-\alpha) = \mu(r-\alpha) - \alpha(r-1),$$

so $\delta_{R2}(\mu) = (N-1)\beta[\mu(r-\alpha) - \alpha(r-1)]/N^2$, which matches the formula in Appendix B.5. At $\mu = \mu_s^{R2} = \alpha(r-1)/(r-\alpha)$, direct substitution gives $\delta_{R2}(\mu_s^{R2}) = 0$, confirming continuity at the R2 cutoff.

Step 4: Additivity. With both corrections in hand, I can now establish that they combine additively.

This is the key structural result: it means the full payoff on any branch of the belief space can be written as the benchmark plus whichever corrections are active, without interaction terms.

The two corrections are independent because they affect disjoint components of H's payoff:

- δ_{R2} operates through Π_H^{prop} (equation (15)), which depends on $V_W^{R2}(\mu)$ but is invariant to the R1 regime.
- δ_{R1} operates through Π_H^W (equation (18)), which depends on the R1 offer type but is invariant to $V_W^{R2}(\mu)$—because the R1 offers to H are pinned at $\beta V_H^{R2}(\theta, \mu'{=}1)$, which is always evaluated at $\mu' = 1 > \mu_s^{R2}$.

Hence the full unanimity payoff is:

$$E[V_H^{R1}(\mu, U)] = E_{\text{bench}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

Step 5: Subtracting majority and collecting terms. The final step subtracts the majority payoff from the unanimity benchmark to obtain D_{base}. This completes the decomposition: the payoff difference $D(\mu)$ on any branch equals D_{base} plus the active corrections, and all three components are affine in μ.

Under majority (Appendix A.1), $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1+(N-1)\alpha) - C_{\text{buy}}]/N^2$ with $C_{\text{buy}} = \beta(q-1)(1-\alpha)$ and $q = \lfloor N/2 \rfloor + 1$. Define $C_{\text{out}} = N(N-1)\alpha$. Then:

$$\begin{aligned} D_{\text{base}}(\mu) &\equiv E_{\text{bench}}(\mu) - \lambda_M V_e(\mu) \\ &= \frac{N\,V_e(\mu) + (N-1)\beta N\alpha r}{N^2} - \frac{N(1+(N-1)\alpha) - C_{\text{buy}}}{N^2} V_e(\mu) \\ &= \frac{V_e(\mu)[N - N - C_{\text{out}} + C_{\text{buy}}] + C_{\text{out}}\beta r}{N^2} \\ &= \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \end{aligned} \tag{24}$$

where the third line uses $N(1+(N-1)\alpha) = N + C_{\text{out}}$ and $(N-1)\beta N\alpha r = C_{\text{out}}\beta r$.

The payoff difference is therefore:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu),$$

with D_{base}, δ_{R2}, δ_{R1} as in equations (24), (23), (22). $\square$

B.5 Proof of Lemma 1 (Conditional payoff dominance)

The payoff difference $D(\mu)$ decomposes into a baseline affine term plus two correction terms, each itself affine. Positivity therefore reduces to checking endpoints. The decomposition is derived in Appendix B.5a.

Let $C_{\text{buy}} \equiv \beta(q-1)(1-\alpha)$ (the vote-buying cost under majority) and $C_{\text{out}} \equiv N(N-1)\alpha$ (the aggregate outside-option cost). Under majority, $E[V_H^{R1}(\mu, M)] = \lambda_M V_e(\mu)$ where $\lambda_M = [N(1+(N-1)\alpha) - C_{\text{buy}}]/N^2$.

Step 1: Decomposition. The decomposition has three components, each affine in μ. Because an affine function that is positive at both endpoints of an interval is positive throughout, it suffices to check a finite number of boundary values.

Define:

$$D_{\text{base}}(\mu) \equiv \frac{(C_{\text{buy}} - C_{\text{out}})V_e(\mu) + C_{\text{out}}\beta r}{N^2}, \tag{25}$$

$$\delta_{R2}(\mu) \equiv \frac{(N-1)\beta[\mu(r-\alpha) - \alpha(r-1)]}{N^2}, \tag{26}$$

$$\delta_{R1}(\mu) \equiv \frac{(N-1)\beta(r-1)(1-\mu)}{N^2}. \tag{27}$$

The difference $D(\mu) \equiv E[V_H^{R1}(\mu, U)] - \lambda_M V_e(\mu)$ decomposes as:

$$D(\mu) = D_{\text{base}}(\mu) + \mathbf{1}\{\mu < \mu_s^{R2}\} \cdot \delta_{R2}(\mu) + \mathbf{1}\{\mu > \mu_s^{R1}\} \cdot \delta_{R1}(\mu).$$

This decomposition holds regardless of the relative ordering of μ_s^{R1} and μ_s^{R2}. The two corrections are additive because they affect independent components of H's payoff: δ_{R2} captures the change in $V_W^{R2}(\mu)$ (which enters through H's proposal offer), while δ_{R1} captures the change in the R1 offer H receives when W proposes (which depends on the R1 regime but not on V_W^{R2}).

D_{base} is the difference under aggressive R1 and conservative R2. The R2 correction vanishes at the screening cutoff: $\delta_{R2}(\mu_s^{R2}) = 0$, ensuring continuity at μ_s^{R2}. The R1 correction satisfies

$\delta_{R1}(\mu) \geq 0$ for all $\mu \leq 1$, with equality only at $\mu = 1$.

Step 2: $D_{\textbf{base}} > 0$ on $[0, 1]$. Since D_{base} is affine in μ—it depends on μ only through $V_e(\mu) = 1 + \mu(r-1)$, which is itself affine—checking positivity at the two endpoints $\mu = 0$ and $\mu = 1$ is both necessary and sufficient.

At $\mu = 1$:

$$D_{\text{base}}(1) = \frac{r[C_{\text{buy}} - C_{\text{out}}(1-\beta)]}{N^2}.$$

Setting $C_{\text{buy}} - C_{\text{out}}(1-\beta) = 0$ and solving for α yields $\alpha = \alpha^*(N, \beta)$. Since $\alpha < \alpha^*$, $D_{\text{base}}(1) > 0$.

At $\mu = 0$: Define $D_I(\mu) \equiv D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, the payoff difference on the low-belief branch. We check $D_I(0)$ rather than $D_{\text{base}}(0)$ directly, because $\delta_{R2}(0) < 0$—so $D_I(0)$ is the tighter endpoint condition. Write $D_I(0) = [\beta(q-1) - \alpha d_0]/N^2$ where $d_0 = N(N-1) - \beta((N-1)^2 r + N - q)$. If $d_0 \leq 0$, then $D_I(0) \geq \beta(q-1)/N^2 > 0$. If $d_0 > 0$, the threshold $\bar{\alpha}_0 = \beta(q-1)/d_0$ satisfies $\bar{\alpha}_0 > \alpha^*$ because $d_0 < d_* \equiv N(N-1) - \beta(N^2 - N - q + 1)$ (since $d_* - d_0 = \beta(N-1)^2(r-1) > 0$). Hence $\alpha < \alpha^* < \bar{\alpha}_0$ implies $D_I(0) > 0$, and therefore $D_{\text{base}}(0) > D_I(0) > 0$.

Conclusion: D_{base} is affine with positive values at both endpoints, hence $D_{\text{base}}(\mu) > 0$ for all $\mu \in [0, 1]$.

Step 3: Assembly. The strategy is to partition the belief space $(0, 1]$ by the two screening cutoffs and verify $D > 0$ on each branch, using the endpoint results from Steps 1–2. On every branch, D is a sum of affine functions and is therefore itself affine, so positivity at branch endpoints implies positivity throughout.

The standard case is $\mu_s^{R2} \leq \mu_s^{R1}$, which partitions $(0, 1]$ into three branches.

Low-belief branch ($\mu < \mu_s^{R2}$): $D = D_I(\mu)$, which is affine. At the left endpoint, $D_I(0) > 0$ (Step 2). At the right endpoint, $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2). Affine with positive endpoints implies $D > 0$ throughout.

Middle branch ($\mu_s^{R2} \leq \mu \leq \mu_s^{R1}$): $D = D_{\text{base}} > 0$ directly from Step 2.

High-belief branch ($\mu > \mu_s^{R1}$): $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Alternative case ($\mu_s^{R1} < \mu_s^{R2}$). The additive decomposition remains valid because the two corrections affect independent components of H's payoff (Step 1). The partition of $(0, 1]$ now has three branches:

Low-belief branch ($\mu < \mu_s^{R1}$): aggressive R1, aggressive R2. $D = D_I(\mu) = D_{\text{base}}(\mu) + \delta_{R2}(\mu)$, which is affine. At $\mu = 0$, $D_I(0) > 0$ (Step 2). At $\mu = \mu_s^{R1}$, D_I is continuous from the left and $D_I(\mu_s^{R1}) > 0$ because D_I is affine with $D_I(0) > 0$ and $D_I(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) > 0$ (Step 2), and $\mu_s^{R1} < \mu_s^{R2}$.

Middle branch ($\mu_s^{R1} \leq \mu < \mu_s^{R2}$): conservative R1, aggressive R2. Both corrections are active: $D = D_{\text{base}} + \delta_{R2} + \delta_{R1}$. This is affine. At μ_s^{R1}, D jumps up by $\delta_{R1}(\mu_s^{R1}) > 0$ relative to the low branch, so $D(\mu_s^{R1}) = D_I(\mu_s^{R1}) + \delta_{R1}(\mu_s^{R1}) > D_I(\mu_s^{R1}) > 0$. At μ_s^{R2}, $\delta_{R2}(\mu_s^{R2}) = 0$, so $D(\mu_s^{R2}) = D_{\text{base}}(\mu_s^{R2}) + \delta_{R1}(\mu_s^{R2}) \geq D_{\text{base}}(\mu_s^{R2}) > 0$. Affine with positive values at both endpoints implies $D > 0$ throughout.

High-belief branch ($\mu \geq \mu_s^{R2}$): conservative R1, conservative R2. $D = D_{\text{base}} + \delta_{R1} \geq D_{\text{base}} > 0$, since $\delta_{R1} \geq 0$.

Combining all branches in either case: $D(\mu) > 0$ for every $\mu \in (0, 1]$. This establishes sufficiency.

Step 4: Necessity. To establish that $\alpha < \alpha^*$ is not merely sufficient but also necessary, I show that the inequality $D(\mu) > 0$ fails at $\mu = 1$ whenever $\alpha \geq \alpha^*$.

At $\mu = 1$, both corrections vanish ($\delta_{R1}(1) = 0$ and $\mu = 1 > \mu_s^{R2}$), so $D(1) = D_{\text{base}}(1) = r[C_{\text{buy}} - C_{\text{out}}(1 - \beta)]/N^2$. Since α^* is the root of $C_{\text{buy}} - C_{\text{out}}(1 - \beta) = 0$, $\alpha \geq \alpha^*$ implies $D(1) \leq 0$, and the strict inequality fails. $\square$

B.6 Proof of Theorem 1 (Dominance of unanimity)

Fix $p \in E_U$, so $V_W^{R1}(p, U) \geq c$. We first show that entry also occurs under majority. Under majority, the budget identity holds exactly: $E[V_H(\mu, M)] + (N-1)V_W(\mu, M) = V_e(\mu)$. Under unanimity, surplus destruction from discounting on the aggressive branch yields $E[V_H(\mu, U)] + (N-1)V_W(\mu, U) \leq V_e(\mu)$. Since Lemma 1 gives $E[V_H(\mu, U)] > E[V_H(\mu, M)]$, it follows that $V_W(\mu, U) < V_W(\mu, M)$ for every $\mu \in (0, 1]$. Hence $V_W^{R1}(p, M) > V_W^{R1}(p, U) \geq c$, so entry occurs under majority at p. In particular, $E_U \subseteq E_M$.

By Lemma 1, $v(p, U) > v(p, M)$. It remains to show that persuasion cannot improve the majority net gain at p. Under majority, whenever entry occurs, $v(\mu, M) = (\lambda_M - \alpha)[1 + (r-1)\mu]$, and whenever entry does not occur, $v(\mu, M) = 0$. Since majority entry is monotone, $E_M = [\tau(M), 1]$.

Consider any Bayes-plausible experiment inducing posteriors (π_s, μ_s) with $\sum_s \pi_s \mu_s = p$. Let $A = \{s : \mu_s \in E_M\}$ be the set of signals after which entry occurs under majority. Then

$$\sum_s \pi_s v(\mu_s, M) = \sum_{s\in A} \pi_s(\lambda_M - \alpha)[1+(r-1)\mu_s] = (\lambda_M - \alpha)\sum_{s\in A}\pi_s + (\lambda_M-\alpha)(r-1)\sum_{s\in A}\pi_s\mu_s.$$

Since $\sum_{s\in A}\pi_s \leq 1$ and $\sum_{s\in A}\pi_s\mu_s \leq \sum_s \pi_s\mu_s = p$, we obtain

$$\sum_s \pi_s v(\mu_s, M) \leq (\lambda_M - \alpha)[1 + (r-1)p] = v(p, M),$$

where the last equality uses $p \in E_M$. Hence cav $v(p, M) = v(p, M)$ for every $p \in E_M$.

Since $\alpha V_e(p)$ is common to both rules, $\Pi^*_H(U, p) > \Pi^*_H(M, p)$ if and only if cav $v(p, U) >$ cav $v(p, M)$. We have cav $v(p, U) \geq v(p, U) > v(p, M) =$ cav $v(p, M)$, which completes the proof. □

B.7 Lemma: Global maximum of V_W^{R1} under unanimity

Lemma 2 (Global maximum of weak-state payoff)**.** $V_W^{R1}(\mu, U) \leq V_W^{R1}(1, U) = r(1-\beta\alpha)/N$ *for every* $\mu \in (0, 1]$, *with equality only at* $\mu = 1$. *Hence if* $E_U \neq \emptyset$, *then* $1 \in E_U$.

Proof. At $\mu = 1$, the equilibrium R1 offer is conservative: $F_1^{con}(1) = r - \beta(r+x)/N - \omega(1) > 0$ while $F_1^{agg}(1) = \beta r(1-\alpha)/N > 0$, and the conservative surplus exceeds the aggressive surplus because $\beta < 1$ implies the full pie exceeds the discounted R2 value. Hence $\bar{V}_W \equiv V_W^{R1}(1, U) = r(1-\beta\alpha)/N$.

We bound every feasible R1 payoff candidate by $\bar{V}_W$, using the full ex ante weak-state payoff (not only the proposer surplus). Let $W_2^L(\mu) = (1-\mu)(1-\alpha)/N$ and $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ denote the R2 continuation values on the low and high branches, $W_2^1 = r(1-\alpha)/N$ the R2 value at $\mu = 1$, and $x = (N-1)\alpha r$. Under a conservative R1 offer (all types accept), a non-proposing weak state receives $\beta W_2^j(\mu)$ whenever another player proposes, giving $V_W^{Cj}(\mu) = F_1^{con,j}(\mu)/N + (N-1)\beta W_2^j(\mu)/N$. Under an aggressive R1 offer ($\theta = 1$ rejects, game proceeds to R2 with posterior one), $V_W^{Aj}(\mu) = F_1^{agg,j}(\mu)/N + \beta W_2^j(\mu)/N + (N-2)[(1-\mu)\beta W_2^j(\mu) + \mu\beta W_2^1]/N$. The equilibrium payoff $V_W^{R1}(\mu, U)$ is one of the four candidates V_W^{CH}, V_W^{AH}, V_W^{CL}, V_W^{AL}, depending on the R1 offer type and the R2 branch.

Conservative R1, high R2 ($\mu \geq \mu_s^{R2}$): $V_W^{CH}(\mu) = [(N+\beta)V_e(\mu) - \beta r(1+N\alpha)]/N^2$, so $\bar{V}_W - V_W^{CH}(\mu) = (N+\beta)(r-1)(1-\mu)/N^2 \geq 0$, with equality only at $\mu = 1$.

Aggressive R1, high R2 ($\mu \geq \mu_s^{R2}$): The proposer surplus $F_1^{agg,H}$ contains a term $(1-\mu)\beta W_2^H(\mu)$ that is quadratic in μ, but this term cancels exactly with the corresponding non-proposer term $(1-\mu)\beta W_2^H(\mu)$ in V_W^{AH}, leaving a result that is affine. After substituting $W_2^H(\mu) = [V_e(\mu) - \alpha r]/N$ and simplifying: $\bar{V}_W - V_W^{AH}(\mu) = [(1-\mu)(r-1) + \mu r(1-\beta)]/N > 0$ for all $\mu \in [0,1]$.

Conservative R1, low R2 ($\mu < \mu_s^{R2}$): Using $W_2^L(\mu) = (1-\mu)(1-\alpha)/N$, $\bar{V}_W - V_W^{CL}(\mu)$ is affine in μ. At $\mu = 0$: $(r-1)(N - \alpha\beta + \beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r-1)(1-\alpha)(N+\beta)/[N^2(r-\alpha)] > 0$. Affine with positive endpoints implies $\bar{V}_W - V_W^{CL} > 0$ on $[0, \mu_s^{R2}]$.

Aggressive R1, low R2 ($\mu < \mu_s^{R2}$): The same quadratic cancellation as in the high-R2 case applies: the $(1-\mu)\beta W_2^L(\mu)$ terms in the proposer and non-proposer components cancel, and $\bar{V}_W - V_W^{AL}(\mu)$ is affine. At $\mu = 0$: $(r-1)(N - \alpha\beta)/N^2 > 0$. At $\mu = \mu_s^{R2}$: $r(r-1)(1-\alpha\beta)/[N(r-\alpha)] > 0$. Hence $\bar{V}_W - V_W^{AL} > 0$ on $[0, \mu_s^{R2}]$.

Since every payoff candidate is strictly below $\bar{V}_W$ for $\mu < 1$, and the equilibrium payoff selects one of these candidates, $V_W^{R1}(\mu, U) < \bar{V}_W$ for all $\mu < 1$. The corollary follows: if $V_W^{R1}(\mu', U) \geq c$ for some μ', then $V_W^{R1}(1, U) \geq V_W^{R1}(\mu', U) \geq c$, so $1 \in E_U$. □ □

B.8 Proof of Theorem 2 (Single-crossing institutional comparison)

If $E_U = \emptyset$, then $v(\mu, U) = 0$ for all μ, so $u = 0 \leq m$ and the dominance set is empty—trivially an upper interval. For the remainder of the proof, assume $E_U \neq \emptyset$. Let $m(p) = \text{cav}\, v(p, M)$ and $u(p) = \text{cav}\, v(p, U)$. Define $a = \inf E_U$.

Preliminary: structure of $m(p)$. Since $E_M = [\tau(M), 1]$ and $v(\mu, M) = (\lambda_M - \alpha)V_e(\mu)$ is affine on E_M (where λ_M is as defined in Appendix B.5), the concavified majority net gain is $m(p) = (\lambda_M - \alpha)V_e(p)$ for $p \geq \tau(M)$ and $m(p) = S_M \cdot p$ for $p < \tau(M)$, where $S_M =$

$(\lambda_M - \alpha)V_e(\tau(M))/\tau(M)$ when $\tau(M) > 0$. If $\tau(M) = 0$, entry occurs at all beliefs and $m(p) = (\lambda_M - \alpha)V_e(p)$ for all p; the slope S_M is not needed. The coefficient $(\lambda_M - \alpha) > 0$ because $(1-\alpha)[N - \beta(q-1)] > 0$. The ratio $m(p)/p$ is weakly decreasing on $(0, 1]$: when $\tau(M) > 0$, it is constant on $(0, \tau(M)]$ and decreasing above $\tau(M)$ (affine function with positive intercept divided by p); when $\tau(M) = 0$, it equals $(\lambda_M - \alpha)(1/p + r - 1)$, which is strictly decreasing.

Part 1: $p \geq a$. Since $V_W^{R1}(\mu, U)$ is piecewise affine with a possible downward jump at μ_s^{R1}, the entry set $E_U = \{\mu : V_W^{R1}(\mu, U) \geq c\}$ is a finite union of closed intervals. Since $E_U \subseteq E_M$, we have $a \geq \tau(M)$. If $p \in E_U$, then $u(p) \geq v(p, U) > v(p, M) = m(p)$ by conditional dominance (Lemma 1), where the last equality holds because $p \geq \tau(M)$ and the concave envelope of an affine function equals the function itself. If $p \notin E_U$ but $p \geq a$, then p lies in a gap (b, d) of E_U with $b, d \in E_U$ (the gap is bounded above because $1 \in E_U$ by Lemma 2; the endpoints belong to E_U because E_U is closed). Since $u(b) > m(b)$ and $u(d) > m(d)$ by the previous argument, and since u is concave, $u(p)$ lies weakly above the chord connecting $(b, u(b))$ and $(d, u(d))$. Since m is affine on $[b, d] \subseteq [\tau(M), 1]$ and the chord is strictly above m at both endpoints, the chord is strictly above m throughout (b, d). Hence $u(p) > m(p)$ for all $p \geq a$.

Part 2: $p < a$. On $[0, a)$, $v(p, U) = 0$. The concave envelope of a function that is zero on $[0, a)$ and nonnegative on $[a, 1]$ is linear from the origin to its first exposed point: there exists $x \geq a$ such that $u(p) = [u(x)/x] \cdot p$ for all $p \in [0, x]$.[10] In particular, $u(p) = S_U \cdot p$ on $[0, a]$ for some slope $S_U \geq 0$ (non-negative because $E_U \neq \emptyset$ implies $v(\mu, U) > 0$ for some μ, so u is strictly positive somewhere on $[a, 1]$).

For $p < a$, $\Delta(p) = u(p) - m(p) = p[S_U - m(p)/p]$. Since $p > 0$, the sign of $\Delta(p)$ equals the sign of $S_U - m(p)/p$. Since $m(p)/p$ is weakly decreasing, $S_U - m(p)/p$ is weakly increasing in p. Hence $\{p \in (0, a) : \Delta(p) > 0\}$ is an upper interval in $(0, a)$.

Combining. Since $\Delta(p) > 0$ for all $p \geq a$ (Part 1) and $\{p \in (0, a) : \Delta(p) > 0\}$ is an upper interval (Part 2), the set $\{p \in (0, 1] : \Delta(p) > 0\}$ is an upper interval in $(0, 1]$. □

[10] Since $u(0) = 0$ and $v = 0$ on $[0, a)$, no point in $(0, a)$ can be an exposed point of the upper envelope. The first such point occurs at some $x \geq a$, and u is affine on $[0, x]$ with $u(0) = 0$.

Appendix C: Robustness to $K > 2$ Types

This appendix shows that the screening mechanism and the majority-linearity result both generalize to $K > 2$ types.

C.1 Setup

The model is identical to Section 4 except that $\theta \in \{0, 1, 2\}$ with $V(0) = 1$, $V(1) = r_1$, $V(2) = r_2$, where $1 < r_1 < r_2$. The prior is $p = (p_0, p_1, p_2) \in \Delta^2$, the 2-simplex. The expected value of cooperation is $V_e(\mu) = \sum_k \mu_k V(\theta_k)$. All other primitives—$N$ players, discount factor β, random proposer, outside option $d_H = \alpha V(\theta)$—are unchanged. The constraint on the outside option becomes $\alpha \in (0, 1/r_2)$, ensuring that $d_H < V(\theta)$ for all types.

C.2 R2 Screening under Unanimity

With K types, the R2 screening problem has a natural generalization. When W makes an offer in R2, it targets a particular type by setting the reservation value equal to $\alpha V(\theta_k)$ for some k. All types θ_j with $j \leq k$ accept (since $\alpha V(\theta_j) \leq \alpha V(\theta_k)$), and all types θ_j with $j > k$ reject. The optimal target balances the probability of acceptance against the cost of overpaying low types.

Claim (R2 screening boundaries for general K). *For K ordered types with values $v_1 < v_2 < \cdots < v_K$, the R2 screening boundary between targeting type k and type $k+1$ is defined by the posterior ratio*

$$\frac{\mu_{k+1}}{\sum_{j \leq k} \mu_j} = \frac{\alpha(v_{k+1} - v_k)}{v_{k+1}(1-\alpha)}.$$

These $K - 1$ conditions define hyperplanes that partition the simplex Δ^{K-1} into K screening regions.

The logic is as follows. The weak proposer's payoff from an offer targeting type k is $\sum_{j \leq k} \mu_j [V(\theta_j) - \alpha V(\theta_k)]$. Equating the payoffs from targeting type k and type $k+1$ yields the indifference condition. The ratio simplifies because the incremental gain from including type $k+1$ (at cost $\alpha V(\theta_{k+1})$) must equal the incremental cost of overpaying all types $j \leq k$ (the additional payment $\alpha[V(\theta_{k+1}) - V(\theta_k)]$ per accepting type). For $K = 2$, this reduces to

$\mu_s^{R2} = \alpha(r-1)/(r-\alpha)$, recovering the main model.

For $K = 3$ with values $(1, r_1, r_2)$, the R2 screening produces three regions separated by two boundaries. The continuation values for H in R2 follow the same logic as the binary case: under a high offer (targeting $\theta = 2$), types $\theta = 0$ and $\theta = 1$ are overpaid; under a medium offer (targeting $\theta = 1$), only $\theta = 0$ is overpaid; under a low offer (targeting $\theta = 0$), all types receive their outside option. The key pattern persists: low types receive strictly more than their outside option whenever W targets a higher type, and the highest type $\theta = 2$ always receives $r_2(1+(N-1)\alpha)/N$ regardless of the screening region.

C.3 Majority: No Screening

Proposition 5 (Linearity of H's Payoff under Majority for General K)**.** *Under majority rule with K types, $E_\theta[V_H^{R1}(\theta, \mu, M)] = \lambda_M V_e(\mu)$, where λ_M is defined as in Theorem 2. No screening cutoff exists for any K.*

Proof. The proof is identical to the $K = 2$ case. Under majority, W never needs H's vote; H receives $\alpha V(\theta)$ in every coalition that forms and $\alpha V(\theta)$ as its outside option. Since H's payoff is $\alpha V(\theta)$ type by type, the expectation is $\alpha V_e(\mu)$ scaled by the bargaining multiplier λ_M/α. The belief μ enters only through $V_e(\mu)$, so the payoff is affine in μ (and hence in V_e) regardless of K. $\square$

Together, the screening claim above and Proposition 5 establish the key asymmetry for general K: unanimity generates $K-1$ screening boundaries (each a potential source of non-convexity), while majority remains linear regardless of K. The binary model is the minimal case that captures this mechanism.

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